

Process Costing

24 problems · workbook pages 8 to 16

What this module actually is

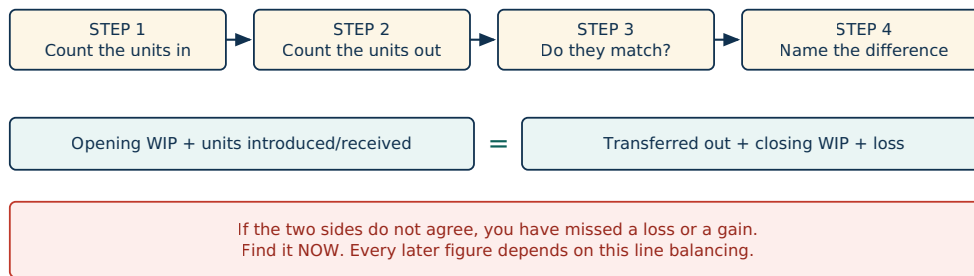
Process costing is used where a product is made in a **continuous chain of stages** — sugar, chemicals, oil, paper, cement. You cannot point at one unit and say “this one cost ₹40”, because every unit goes through the same pipe. So we do the only thing possible: **total the cost of a process and divide by the units it produced**. That single sentence is the whole module.

Everything else in these 24 problems is one of four complications layered on top of that idea:

Complication	What changes	Problems
Nothing — plain chain	Total each process, divide by units, carry the total forward to the next process.	1-4
Some units are lost	Part of the input disappears (evaporation, spoilage). Expected loss is <i>normal</i> and its cost is absorbed by the good units. Anything worse is <i>abnormal loss</i> ; anything better is <i>abnormal gain</i> . Both are taken out of the process at the good-unit rate.	5-9
Two or more products emerge	One process throws out several saleable products. The joint cost has to be split between them — usually by working <i>backwards</i> from selling price.	10-14
Each process adds profit	A process sells to the next at cost + profit. The profit inside unsold stock is not real yet, so a <i>reserve for unrealised profit</i> must be removed.	15-16
Some units are half-finished	Closing work-in-progress is 60% done. You convert part-finished units into <i>equivalent</i> finished units before dividing.	17-24

The one habit that solves every problem in this module

Before you write a single figure, do this on your rough sheet. It takes forty seconds and it is the difference between a full-marks answer and a mess.



The units reconciliation. Write it for every process, every time.

THE THREE RATE FORMULAS — THIS IS ALL THE ARITHMETIC THERE IS

$$\text{Cost per unit (no loss)} = \frac{\text{Total cost of the process}}{\text{Units produced}}$$

$$\text{Cost per good unit (with loss)} = \frac{\text{Total cost} - \text{Scrap value of normal loss}}{\text{Input units} - \text{Normal loss units}}$$

Learn this one properly. Normal loss is removed from BOTH the top and the bottom. Abnormal loss and abnormal gain are then valued at this same rate — never at a different one.

$$\text{Cost per unit (part-finished stock)} = \frac{\text{Cost of that element}}{\text{Equivalent units of that element}}$$

Material, labour and overhead each get their own rate, because they are each at a different stage of completion.

Q1 Three processes, overhead on wages

Plain chain · p.8

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

There are three distinct processes A, B and C. During the four-week period 1,000 units are produced and the following information is available.

Particulars	Process A	Process B	Process C
Direct materials	2,000	1,000	—
Direct wages	1,500	700	800
Direct expenses	300	100	—

The indirect production cost were ₹4,500 and these are to be apportioned on the basis of direct wage cost. Prepare the process accounts.

READ THE QUESTION FIRST — WHAT TO SPOT

Three things to notice, and they decide the whole answer:

- **“1,000 units are produced”** — one figure for all three processes. So there is **no loss anywhere**. 1,000 units go in, 1,000 come out of every process. That means no normal loss working, no abnormal loss — the easiest kind.
- **“apportioned on the basis of direct wage cost”** — the question hands you the basis. You do not have to guess it. Whenever a question gives you a lump overhead and a basis, your first working note is always the same: find the ratio, then split.
- **Process C has no materials and no direct expenses**. Do not panic and do not leave the row out — write a dash. An examiner wants to see that you looked and found nothing.

THE METHOD FOR THIS TYPE — ALWAYS THESE STEPS

1. Add up the given basis across all processes (here: total direct wages).
2. Overhead rate = Total indirect cost ÷ Total basis. Keep it as a plain number or a percentage — whichever divides cleanly.
3. Multiply each process's basis by that rate. **Check the three pieces add back to the original lump sum** before you go on.
4. Open Process A. Debit its own four costs. Total it. That total transfers out to B.
5. Open Process B. **First debit is the transfer from A**, then B's own costs. Total, transfer to C.
6. Open Process C the same way. Its total goes to Finished Stock.
7. Divide each process total by 1,000 to show cost per unit. The last one is the answer the examiner is looking for.

WORKING NOTES

W1 Apportionment of indirect production cost

Total direct wages = 1,500 + 700 + 800 = **₹3,000**

Overhead rate = ₹4,500 ÷ ₹3,000 = **1.5 times the wages** (i.e. 150% of direct wages)

Process	Direct wages	× 1.5	Overhead absorbed
A	1,500	1,500 × 1.5	2,250
B	700	700 × 1.5	1,050
C	800	800 × 1.5	1,200
Total	3,000		4,500

Cross-check: 2,250 + 1,050 + 1,200 = 4,500 equals the indirect cost given. If this does not tally, stop and re-divide.

Process A Account

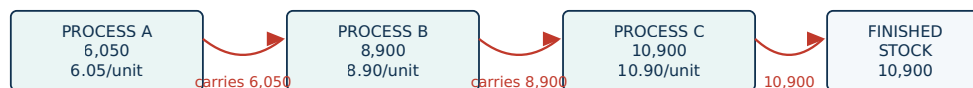
Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Direct materials	1,000	—	2,000	By Transfer to Process B 6,050 ÷ 1,000 units = ₹6.05 per unit	1,000	6.05	6,050
To Direct wages	—	—	1,500				
To Direct expenses	—	—	300				
To Indirect production cost W1: 1,500 × 1.5	—	—	2,250				
Total	1,000	—	6,050	Total	1,000	—	6,050

Process B Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Transfer from Process A ← total of Process A a/c	1,000	6.05	6,050	By Transfer to Process C $8,900 \div 1,000 \text{ units} = ₹8.90 \text{ per unit}$	1,000	8.90	8,900
To Direct materials	—	—	1,000				
To Direct wages	—	—	700				
To Direct expenses	—	—	100				
To Indirect production cost W1: 700×1.5	—	—	1,050				
Total	1,000	—	8,900	Total	1,000	—	8,900

Process C Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Transfer from Process B ← total of Process B a/c	1,000	8.90	8,900	By Transfer to Finished Stock $10,900 \div 1,000 = ₹10.90 \text{ per unit}$	1,000	10.90	10,900
To Direct materials	—	—	—				
To Direct wages	—	—	800				
To Direct expenses	—	—	—				
To Indirect production cost W1: 800×1.5	—	—	1,200				
Total	1,000	—	10,900	Total	1,000	—	10,900



Each box adds only its OWN cost; the arrow carries the whole running total forward.

The chain. Notice the cost per unit can only go up as you move right.

THE TRAP / WHERE MARKS ARE LOST

- Students write Process B's cost per unit as $2,850 \div 1,000 = ₹2.85$, using only B's own costs. Wrong. The unit leaving B has A's cost inside it too. **The transfer-in is a debit like any other.**
- Forgetting to show **units** in the account. Marks are given for the units column, and it is free — every figure is 1,000.
- Apportioning the overhead on the wrong basis (materials, or equally between three processes). The question said direct wage cost. Read it, underline it, use it.

FINAL ANSWER

Cost per unit — Process A	₹ 6.05
Cost per unit — Process B	₹ 8.90
Cost per unit — Process C (final)	₹ 10.90
Total cost of 1,000 units	₹ 10,900

State the final cost per unit in words as your closing line: "The cost of production is ₹10,900 for 1,000 units, i.e. ₹10.90 per unit."

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

The following details show the cost of the three processes of manufacturing. The production of each process is passed on to the next till completion.

Particulars	Process A	Process B	Process C
Wages and materials	60,800	24,000	58,500
Works overheads	11,200	10,500	12,000
Production in units	72,000	75,000	96,000
Opening stock (units from preceding process on 1st July)	—	8,000	33,000
Closing stock (units from preceding process on 31st July)	—	2,000	11,000

READ THE QUESTION FIRST — WHAT TO SPOT

This problem looks like Q1 but it is not, and one phrase tells you so: “**units from preceding process**”.

- The opening and closing stock in Process B is **not half-finished work**. It is a pile of finished A-units waiting to be worked on. So it is valued at **A's rate**, not B's. Same for C — its stock is B-units, valued at B's rate.
- Because there is stock lying on both sides, the units B actually *consumed* are not the units it *received*. You must work the consumption out.
- “Production in units” is the **output** of that process. Divide by this figure to get the rate — never by the units consumed.

THE METHOD FOR THIS TYPE — ALWAYS THESE STEPS

1. **Process A first** — it has no stock, so rate = total cost ÷ production.
2. For B and C, find the units consumed: **Opening stock + Received from previous process – Closing stock**.
3. Value those consumed units at the **previous process's rate**. That is the transfer-in cost.
4. Add the process's own wages, materials and works overheads.
5. Rate for this process = that grand total ÷ **its own production in units**.
6. Now write the account. Debit side: opening stock, transfer-in, own costs. Credit side: closing stock (at the *previous* rate) and transfer-out (at *this* rate). The two sides must balance to the rupee.

WORKING NOTES

W1 Process A rate (no stock, so straight division)

Total cost = 60,800 + 11,200 = **₹72,000**

Rate = 72,000 ÷ 72,000 units = **₹1.00 per unit**

W2 Units consumed by Process B and Process C

Particulars	Process B	Process C
Opening stock of previous-process units	8,000	33,000
Add: Received from previous process	72,000	75,000
Units available	80,000	1,08,000

Particulars	Process B	Process C
Less: Closing stock of previous-process units	(2,000)	(11,000)
Units consumed in the process	78,000	97,000

Received by B = A's production 72,000. Received by C = B's production 75,000.

W3 Rate for Process B

Cost of units consumed = $78,000 \times ₹1.00 = 78,000$
Add own cost = $24,000 + 10,500 = 34,500$
Total = **₹1,12,500**
Rate = $1,12,500 \div 75,000$ units produced = **₹1.50 per unit**

W4 Rate for Process C

Cost of units consumed = $97,000 \times ₹1.50 = 1,45,500$
Add own cost = $58,500 + 12,000 = 70,500$
Total = **₹2,16,000**
Rate = $2,16,000 \div 96,000$ units produced = **₹2.25 per unit**

Process A Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Wages and materials	72,000	—	60,800	By Transfer to Process B W1: rate ₹1.00	72,000	1	72,000
To Works overheads	—	—	11,200				
Total	72,000	—	72,000	Total	72,000	—	72,000

Process B Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Opening stock 8,000 A-units @ ₹1.00 ← A's rate	8,000	1	8,000	By Closing stock 2,000 A-units @ ₹1.00 — still A's rate	2,000	1	2,000
To Transfer from Process A	72,000	1	72,000	By Transfer to Process C W3: 75,000 @ ₹1.50	75,000	1.50	1,12,500
To Wages and materials	—	—	24,000	(3,000 units consumed but not yielded — process shrinkage, no scrap value given)	3,000	—	—
To Works overheads	—	—	10,500				
Total	80,000	—	1,14,500	Total	80,000	—	1,14,500

Process C Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Opening stock <i>33,000 B-units @ ₹1.50 ← B's rate</i>	33,000	1.50	49,500	By Closing stock <i>11,000 B-units @ ₹1.50</i>	11,000	1.50	16,500
To Transfer from Process B	75,000	1.50	1,12,500	By Transfer to Finished Stock <i>W4: 96,000 @ ₹2.25</i>	96,000	2.25	2,16,000
To Wages and materials	—	—	58,500	<i>(1,000 units shrinkage)</i>	1,000	—	—
To Works overheads	—	—	12,000				
Total	1,08,000	—	2,32,500	Total	1,08,000	—	2,32,500



Why Process B's stock is valued at A's rate, not B's.

THE TRAP / WHERE MARKS ARE LOST

- Valuing B's opening/closing stock at B's own rate of ₹1.50. It has not been through B yet — it is worth ₹1.00. This single error breaks both sides of the account.
- Dividing by units consumed (78,000) instead of units produced (75,000). The rate must describe a unit that came out.
- Panicking when the units do not balance. Here 78,000 are consumed but only 75,000 emerge. That gap is real shrinkage. No scrap value is given, so it carries no cost — simply show it and move on. Do not invent a normal-loss working the question never asked for.

FINAL ANSWER

Cost per unit — Process A	₹ 1.00
Cost per unit — Process B	₹ 1.50
Cost per unit — Process C (final)	₹ 2.25
Cost transferred to Finished Stock	₹ 2,16,000

Both B and C accounts balance exactly — 1,14,500 and 2,32,500. Always show the totals; a balancing account is self-proving.

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

From the following information relating to an article which undergoes three processes for manufacture, show the cost of each process and the cost per article.

Particulars	Process A	Process B	Process C
Materials consumed	37,500	12,500	50,000
Labour	20,000	50,000	15,000
Direct expenses	6,500	18,000	6,250

The indirect expenses amounted to ₹21,250 and the number of articles produced during the month was 240 units.

READ THE QUESTION FIRST — WHAT TO SPOT

- Same shape as Q1 — one output figure (240 articles), so **no losses**.
- **The basis for splitting the indirect expenses is NOT stated.** This is deliberate. The accepted convention when no basis is given is to apportion on **direct labour**. Say so in one line in your answer — “indirect expenses apportioned on the basis of direct labour in the absence of any other basis” — and you protect the marks.
- Test whether your assumed basis is right by checking the split comes out in round figures. Here labour gives exactly 25%. That is the examiner confirming your choice.
- Two answers are wanted: **cost of each process** and **cost per article**. Give both, clearly labelled.

THE METHOD FOR THIS TYPE — ALWAYS THESE STEPS

1. Total the chosen basis — labour: 20,000 + 50,000 + 15,000.
2. Rate = 21,250 ÷ that total. Express as a percentage if it is clean.
3. Split, then cross-check the pieces add back to 21,250.
4. Build the three process accounts in a chain, transferring the running total.
5. Divide each process total by 240 to get cost per article at that stage.
6. Final line: cost per article = Process C total ÷ 240.

WORKING NOTES

W1 Apportionment of indirect expenses (basis: direct labour)

Total labour = 20,000 + 50,000 + 15,000 = **₹85,000**

Rate = 21,250 ÷ 85,000 = 0.25 = **25% of direct labour**

Process	Labour	× 25%	Indirect expenses
A	20,000	20,000 × 0.25	5,000
B	50,000	50,000 × 0.25	12,500
C	15,000	15,000 × 0.25	3,750
Total	85,000		21,250

Cross-check: 5,000 + 12,500 + 3,750 = 21,250 — and every figure is a round number, which confirms labour was the intended basis.

Statement of Cost of Each Process and Cost per Article

Particulars	Process A	Process B	Process C
Cost of previous process (transferred in)	—	69,000	1,62,000
Materials consumed	37,500	12,500	50,000
Labour	20,000	50,000	15,000
Direct expenses	6,500	18,000	6,250
Indirect expenses (W1)	5,000	12,500	3,750
Total cost of the process	69,000	1,62,000	2,37,000
Cost per article (÷ 240)	287.50	675.00	987.50

Process A Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Materials consumed	240	—	37,500	By Transfer to Process B $69,000 \div 240 = ₹287.50$	240	287.50	69,000
To Labour	—	—	20,000				
To Direct expenses	—	—	6,500				
To Indirect expenses W1	—	—	5,000				
Total	240	—	69,000	Total	240	—	69,000

Process B Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Transfer from Process A	240	287.50	69,000	By Transfer to Process C $1,62,000 \div 240 = ₹675.00$	240	675	1,62,000
To Materials consumed	—	—	12,500				
To Labour	—	—	50,000				
To Direct expenses	—	—	18,000				
To Indirect expenses W1	—	—	12,500				
Total	240	—	1,62,000	Total	240	—	1,62,000

Process C Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Transfer from Process B	240	675	1,62,000	By Transfer to Finished Stock $2,37,000 \div 240 = ₹987.50$	240	987.50	2,37,000
To Materials consumed	—	—	50,000				
To Labour	—	—	15,000				
To Direct expenses	—	—	6,250				
To Indirect expenses W1	—	—	3,750				
Total	240	—	2,37,000	Total	240	—	2,37,000

WHY THIS WORKS

The statement and the three accounts say exactly the same thing. If the question says “show the cost of each process” the statement alone is enough; if it says “prepare process accounts” you must draw the accounts. When in doubt give the statement first — it is faster to write and it makes the accounts almost automatic.

FINAL ANSWER

Cost of Process A ₹ 69,000

Cost of Process B ₹ 1,62,000

Cost of Process C ₹ 2,37,000

Cost per article ₹ 987.50

Q4 By-product and residue

Chain + by-product · p.8-9

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

A particular brand of chemical passed through three processes during the week ended 15th January. 600 tins were produced. The following information is disclosed by each process.

Particulars	Process A	Process B	Process C
Materials	4,000	2,000	1,500
Labour expenses	3,000	2,500	2,300
Direct expenses	6,000	200	500
Cost of tins	—	—	2,030
Packing charges	—	—	325

The indirect expenses were ₹1,600. The by-product is sold for ₹240 (Process B). The residue was sold for ₹125.50 (Process C).

Prepare the process A/c with cost of production of the product per tin.

READ THE QUESTION FIRST — WHAT TO SPOT

- Again 600 tins throughout — **no process loss**.
- Two new items: a **by-product sold for 240** and a **residue sold for 125.50**. Both are **deducted** from the cost of the process in which they arise. They are not income and they do not go to Profit & Loss — they reduce the cost that the good output has to carry.
- “Cost of tins” and “packing charges” belong to Process C only. They are ordinary debits — do not treat them as anything special.
- The indirect expense basis is again not stated, so apportion on labour and say so.

THE METHOD FOR THIS TYPE — ALWAYS THESE STEPS

1. Apportion the indirect expenses on labour (state the assumption).
2. Build Process A normally.
3. In Process B, debit everything, then **credit the by-product sale of 240**. The amount that transfers to C is the total *after* that deduction.
4. In Process C, debit everything including tins and packing, then **credit the residue sale of 125.50**.
5. Cost of production = final balance of Process C. Divide by 600 tins.

WORKING NOTES

W1 Apportionment of indirect expenses (basis: labour)

Total labour = 3,000 + 2,500 + 2,300 = ₹7,800

Rate = 1,600 ÷ 7,800 = 0.205128 per rupee of labour

Process	Labour	Working	Indirect expenses
A	3,000	$3,000 \times 0.205128$	615.38
B	2,500	$2,500 \times 0.205128$	512.82
C	2,300	$2,300 \times 0.205128$	471.80
Total	7,800		1,600.00

The rate does not divide neatly here, so carry two decimals and force the last figure to make the total exactly 1,600. Write “difference of ₹0.01 adjusted in Process C” if you like — examiners accept it and it shows you checked.

Process A Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Materials	600	—	4,000	By Transfer to Process B	600	22.69	13,615.38
To Labour expenses	—	—	3,000				
To Direct expenses	—	—	6,000				
To Indirect expenses W1	—	—	615.38				
Total	600	—	13,615.38	Total	600	—	13,615.38

Process B Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Transfer from Process A	600	22.69	13,615.38	By Sale of by-product deducted from cost, NOT income	—	—	240
To Materials	—	—	2,000	By Transfer to Process C $18,828.20 - 240 = 18,588.20$	600	30.98	18,588.20
To Labour expenses	—	—	2,500				
To Direct expenses	—	—	200				
To Indirect expenses W1	—	—	512.82				
Total	600	—	18,828.20	Total	600	—	18,828.20

Process C Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Transfer from Process B	600	30.98	18,588.20	By Sale of residue deducted from cost	—	—	125.50
To Materials	—	—	1,500	By Cost of production transferred $25,715.00 - 125.50 = 25,589.50$	600	42.65	25,589.50
To Labour expenses	—	—	2,300				
To Direct expenses	—	—	500				
To Cost of tins	—	—	2,030				
To Packing charges	—	—	325				
To Indirect expenses W1	—	—	471.80				
Total	600	—	25,715	Total	600	—	25,715

THE TRAP / WHERE MARKS ARE LOST

- Adding the by-product sale to the credit of Profit & Loss instead of deducting it inside the process. In a **by-product of small value** the standard treatment is to credit the process. That is what “(Process B)” in the question is telling you.
- Deducting the residue from Process B or from the total instead of from Process C. The question names the process for each — use it.
- Rounding the per-tin figure too early. Carry the paise through and round only at the last line.

FINAL ANSWER

Total cost of production ₹ **25,589.50**

Number of tins **600**

Cost of production per tin ₹ **42.65**

Closing line: “Cost of production is ₹25,589.50 for 600 tins, i.e. ₹42.65 per tin.”

Q5 Normal loss and abnormal loss

Loss & gain · p.9

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

Prepare a Process Account and Abnormal Loss Account from the following information.

Particulars	
Input of raw material	1,000 units @ ₹20 per unit
Direct material	₹4,200
Direct wages	₹6,000
Production overheads	₹6,000
Actual output transferred to Process II	900 units
Normal loss	5%
Value of scrap per unit	₹8

READ THE QUESTION FIRST — WHAT TO SPOT

This is the first problem with a loss, and it is the template for Q5–Q9. Four figures decide everything:

- **Input 1,000 units.** Normal loss is a percentage *of input* — so 5% of 1,000 = 50 units. Not 5% of output.
- **Expected output = 1,000 – 50 = 950 units.** Write this down. It is the figure you compare the actual against.
- **Actual output is 900**, which is *less* than expected 950. So 50 units have gone missing beyond what was allowed — that is **abnormal loss**. If actual had been more than 950 it would have been abnormal gain.
- **Scrap is worth 8 per unit.** Normal loss units are sold as scrap, so they do carry a value — and that value is removed from the cost before you divide.

THE METHOD FOR THIS TYPE — ALWAYS THESE STEPS

1. Total the debits — every cost that entered the process.
2. Normal loss in **units** = given % × input units. Normal loss in **rupees** = those units × scrap rate.
3. Expected output = Input – Normal loss units.
4. Compare actual output with expected output. Shortfall = abnormal loss; excess = abnormal gain.
5. **Cost per good unit** = (Total cost – Scrap value of normal loss) ÷ (Input units – Normal loss units). Show this fraction in full.
6. Value the transfer-out AND the abnormal loss at that **same** rate.
7. Write the process account — it must balance. Then open the Abnormal Loss account: debit the process value, credit the scrap realised, and the balance goes to Profit & Loss.

WORKING NOTES

W1 Total cost debited to the process

Raw material 1,000 × ₹20 = 20,000
Direct material = 4,200
Direct wages = 6,000
Production overheads = 6,000
Total = ₹36,200

W2 The units reconciliation

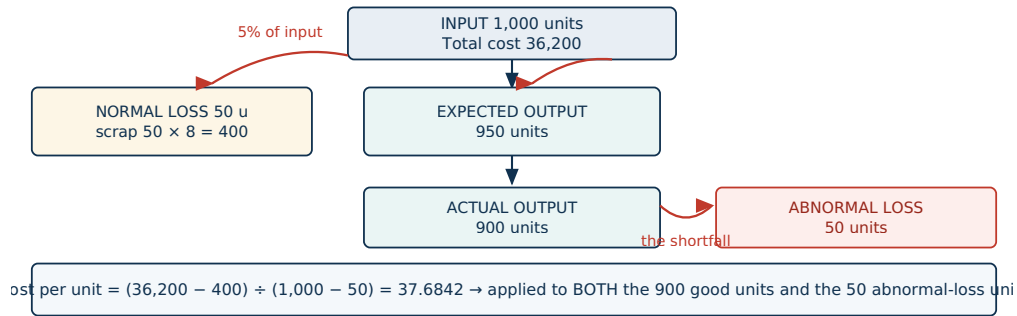
Normal loss = 5% of 1,000 = **50 units**, scrap value 50 × ₹8 = **₹400**
Expected output = 1,000 – 50 = **950 units**
Actual output = 900 units
Abnormal loss = 950 – 900 = **50 units**

W3 Cost per good unit — the formula that matters most

$$\text{₹36,200} - \text{₹400} \div 1,000 - 50 = \text{₹35,800} \div 950 = \text{₹37.6842 per unit}$$

Normal loss is taken out of the top (its scrap value) and out of the bottom (its units). That is how the cost of the normal loss gets loaded onto the good units.

Transferred out 900 × 37.6842 = **₹33,915.79**
Abnormal loss 50 × 37.6842 = **₹1,884.21**



Normal loss is expected and free. Abnormal loss is the shortfall against expectation.

Process I Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Raw material	1,000	20	20,000	By Normal loss 50 units × ₹8 scrap	50	8	400
To Direct material	—	—	4,200	By Abnormal loss W3: 50 × 37.6842	50	37.68	1,884.21
To Direct wages	—	—	6,000	By Transfer to Process II W3: 900 × 37.6842	900	37.68	33,915.79
To Production overheads	—	—	6,000				
Total	1,000	—	36,200	Total	1,000	—	36,200

Abnormal Loss Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Process I A/c ← from the process account	50	37.68	1,884.21	By Sale of scrap 50 units × ₹8 — abnormal loss units are also sold as scrap	50	8	400
Total	50	—	1,884.21	By Profit & Loss A/c balance — the real loss to the business	—	—	1,484.21
				Total	50	—	1,884.21

Normal Loss Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Process I A/c	50	8	400	By Cost Ledger Control A/c (cash from scrap)	50	8	400
Total	50	—	400	Total	50	—	400

THE TRAP / WHERE MARKS ARE LOST

- Dividing by 1,000 instead of 950. You would get ₹35.80 and every later figure would be wrong. **The denominator is always input minus normal loss units.**
- Forgetting to deduct the 400 scrap value from the numerator.
- Valuing the abnormal loss at the scrap rate of 8. No — abnormal loss is valued at the **good-unit rate**. It only realises 8 when sold, and the difference is the loss you take to Profit & Loss.
- Taking the whole 1,884.21 to Profit & Loss. Only the **net** figure of 1,484.21 goes there, after crediting the 400 the scrap actually fetched.

FINAL ANSWER

Cost per good unit **₹ 37.6842**

Transferred to Process II (900 units)	₹ 33,915.79
Abnormal loss (50 units, at cost)	₹ 1,884.21
Scrap realised on abnormal loss	₹ 400.00
Net abnormal loss charged to Profit & Loss	₹ 1,484.21

Note how the Process account, the Abnormal Loss account and the Normal Loss account all close cleanly. Show all three unless the question asks for fewer.

Q6 Loss, gain and neither — all in one

Loss & gain · p.9

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

The product of a company passes through 3 distinct processes. The following information is obtained from the accounts for the month ending January 31, 2024.

Particulars	Process A	Process B	Process C
Direct material	7,800	5,940	8,886
Direct wages	6,000	9,000	12,000
Production overheads	6,000	9,000	12,000

3,000 units @ ₹3 each were introduced to Process I. There was no stock of materials or work in progress. The output of each process passes directly to the next process and finally to Finished Stock A/c. The following additional data is obtained:

Process	Output	Normal loss %	Realisable value of scrap
Process 1	2,850	5%	2
Process 2	2,520	10%	4
Process 3	2,250	15%	5

Prepare Process Cost Account, Normal Loss Account and Abnormal Gain or Loss Account.

READ THE QUESTION FIRST — WHAT TO SPOT

This is **the** model problem of the module. It contains all three possibilities in one question, which is exactly why examiners like it.

- **Process 1 ends with no abnormal item at all** — actual output equals expected. Do not force an abnormal figure just because the other processes have one.
- **Process 2 has an abnormal loss** (output less than expected).
- **Process 3 has an abnormal gain** (output MORE than expected). An abnormal gain is entered on the **debit** side of the process account — the opposite side from a loss. This single fact is the most commonly dropped mark in the whole module.
- Normal loss % is always applied to **the units entering that process**, which is the previous process's output — not the original 3,000.

THE METHOD FOR THIS TYPE — ALWAYS THESE STEPS

1. Work the processes strictly in order. Each one needs the previous one's output *and* its rate.
2. For each process: total the debits, compute normal loss units and scrap value, find expected output, compare with actual.
3. Rate = (Total cost – normal loss scrap) ÷ (Input units – normal loss units).

4. **Loss** → credit the process with the shortfall at that rate. **Gain** → debit the process with the excess at that rate.
5. Open one Normal Loss A/c for all three processes together. Debit each process's scrap value into it.
6. If there is an abnormal gain, the normal loss did not fully happen — so **credit Normal Loss A/c and debit Abnormal Gain A/c** with the gain units × that process's scrap rate.
7. Close Abnormal Loss and Abnormal Gain to Profit & Loss.

WORKING NOTES

W1 Process 1

Total cost = $(3,000 \times ₹3 = 9,000) + 7,800 + 6,000 + 6,000 = ₹28,800$

Normal loss = 5% of 3,000 = **150 units**; scrap $150 \times ₹2 = ₹300$

Expected output = $3,000 - 150 = 2,850$ units. Actual = 2,850 units.

Difference = NIL — no abnormal loss or gain in Process 1.

$$\text{Rate} = \frac{28,800 - 300}{3,000 - 150} = \frac{28,500}{2,850} = ₹10.00 \text{ per unit}$$

W2 Process 2

Total cost = $(2,850 \times ₹10 = 28,500) + 5,940 + 9,000 + 9,000 = ₹52,440$

Normal loss = 10% of **2,850** = **285 units**; scrap $285 \times ₹4 = ₹1,140$

Expected output = $2,850 - 285 = 2,565$ units. Actual = 2,520 units.

Abnormal LOSS = 2,565 - 2,520 = 45 units

$$\text{Rate} = \frac{52,440 - 1,140}{2,850 - 285} = \frac{51,300}{2,565} = ₹20.00 \text{ per unit}$$

Abnormal loss $45 \times 20 = ₹900$

Transfer to Process 3 $2,520 \times 20 = ₹50,400$

W3 Process 3

Total cost = $(2,520 \times ₹20 = 50,400) + 8,886 + 12,000 + 12,000 = ₹83,286$

Normal loss = 15% of **2,520** = **378 units**; scrap $378 \times ₹5 = ₹1,890$

Expected output = $2,520 - 378 = 2,142$ units. Actual = 2,250 units.

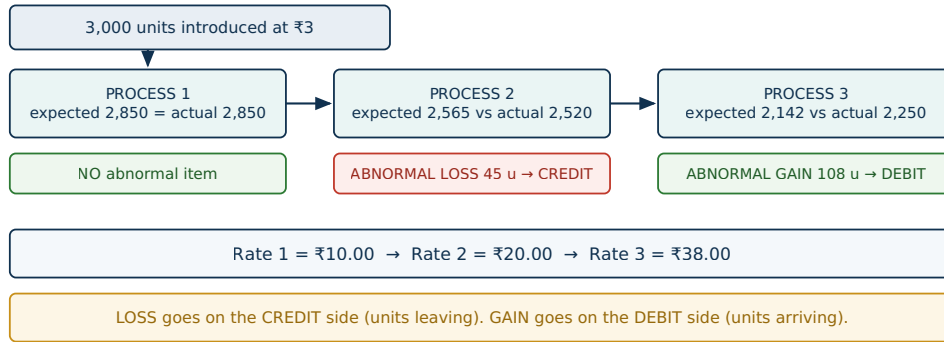
Abnormal GAIN = 2,250 - 2,142 = 108 units (actual is MORE than expected)

$$\text{Rate} = \frac{83,286 - 1,890}{2,520 - 378} = \frac{81,396}{2,142} = ₹38.00 \text{ per unit}$$

Note the rate is still built on EXPECTED output (2,142), not on the actual 2,250. The gain is then valued at this rate and added back.

Abnormal gain $108 \times 38 = ₹4,104$ (debit side)

Transfer to Finished Stock $2,250 \times 38 = ₹85,500$



One question, all three outcomes. Learn this picture and Q5 to Q9 all become the same problem.

Process 1 Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Units introduced	3,000	3	9,000	By Normal loss $150 \text{ u} \times ₹2$	150	2	300
To Direct material	—	—	7,800	By Transfer to Process 2 $W1: 2,850 \times ₹10$	2,850	10	28,500
To Direct wages	—	—	6,000				
To Production overheads	—	—	6,000				
Total	3,000	—	28,800	Total	3,000	—	28,800

Process 2 Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Transfer from Process 1	2,850	10	28,500	By Normal loss $285 \text{ u} \times ₹4$	285	4	1,140
To Direct material	—	—	5,940	By Abnormal loss $W2: 45 \times ₹20$	45	20	900
To Direct wages	—	—	9,000	By Transfer to Process 3 $W2: 2,520 \times ₹20$	2,520	20	50,400
To Production overheads	—	—	9,000				
Total	2,850	—	52,440	Total	2,850	—	52,440

Process 3 Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Transfer from Process 2	2,520	20	50,400	By Normal loss $378 \text{ u} \times ₹5$	378	5	1,890
To Direct material	—	—	8,886	By Transfer to Finished Stock $W3: 2,250 \times ₹38$	2,250	38	85,500
To Direct wages	—	—	12,000				
To Production overheads	—	—	12,000				
To Abnormal gain $W3: 108 \times ₹38$ — DEBIT side	108	38	4,104				
Total	2,628	—	87,390	Total	2,628	—	87,390

Normal Loss Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Process 1 A/c	150	2	300	By Abnormal Gain A/c <i>108 u × ₹5 — the Process 3 normal loss that never happened</i>	108	5	540
To Process 2 A/c	285	4	1,140	By Cost Ledger Control A/c (scrap actually sold)	705	—	2,790
To Process 3 A/c	378	5	1,890				
Total	813	—	3,330	Total	813	—	3,330

Abnormal Loss Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Process 2 A/c	45	20	900	By Cost Ledger Control A/c <i>45 u × ₹4 scrap</i>	45	4	180
Total	45	—	900	By Profit & Loss A/c balance = the real loss	—	—	720
				Total	45	—	900

Abnormal Gain Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Normal Loss A/c <i>108 u × ₹5 — scrap income forgone</i>	108	5	540	By Process 3 A/c	108	38	4,104
To Profit & Loss A/c <i>balance = the real gain</i>	—	—	3,564				
Total	108	—	4,104	Total	108	—	4,104

THE TRAP / WHERE MARKS ARE LOST

- **Putting the abnormal gain on the credit side.** Think about it physically: extra units have *arrived* in the process, so they must be debited like any other input. If your Process 3 account will not balance, this is why.
- Applying the 15% normal loss to 3,000 instead of 2,520. Each percentage applies to that process's own input.
- Forgetting the entry **Abnormal Gain A/c → Normal Loss A/c 540**. Because 108 units did not perish, the company also did not earn scrap on them — so the gain must be reduced by $108 \times ₹5$. Without this the Normal Loss account will not close.
- Using the actual output 2,250 as the denominator for the rate. The denominator is always **input minus normal loss** = 2,142.

FINAL ANSWER

Cost per unit — Process 1	₹ 10.00
Cost per unit — Process 2	₹ 20.00
Cost per unit — Process 3	₹ 38.00
Abnormal loss (Process 2, 45 units)	₹ 900
Abnormal gain (Process 3, 108 units)	₹ 4,104
Transferred to Finished Stock (2,250 units)	₹ 85,500

Net loss to P&L on abnormal loss ₹ 720

Net gain to P&L on abnormal gain ₹ 3,564

Every one of the six accounts closes exactly. If yours do not, check the abnormal gain is on the debit side and that the 540 entry is present.

Q7 Abnormal gain only

Loss & gain · p.10

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

Product A is obtained after it passes through three distinct processes. 2,000 kgs of material at ₹5 per kg were issued to Process 1, direct wages amounted to ₹900 and other production overheads incurred was ₹500. Normal loss is estimated at 10% of input. The wastage is sold at ₹3 per kg. The actual output is 1,850 kgs. Prepare Process 1 A/c, Normal Loss A/c and Abnormal Gain A/c.

READ THE QUESTION FIRST — WHAT TO SPOT

- Only **Process 1** is asked for. The question mentions three processes to set the scene — ignore the other two. Answer exactly what is asked.
- Expected output = 2,000 – 200 = 1,800 kg. Actual is **1,850 kg — higher**. The question even names the account it wants: **Abnormal Gain A/c**. That is your confirmation before you calculate anything.
- Because there is a gain, remember the two-part consequence: debit the process with the gain at cost, and transfer the forgone scrap value to Normal Loss A/c.

THE METHOD FOR THIS TYPE — ALWAYS THESE STEPS

1. Total the debits.
2. Normal loss = 10% of 2,000 = 200 kg; scrap value = 200 × 3.
3. Expected output 1,800 kg vs actual 1,850 kg → abnormal gain 50 kg.
4. Rate = (11,400 – 600) ÷ (2,000 – 200).
5. Debit the process with 50 kg at that rate; credit the transfer of 1,850 kg at that rate.
6. Normal Loss A/c: debit 200 kg at scrap rate; credit 50 kg to Abnormal Gain A/c; the rest is scrap sold.
7. Abnormal Gain A/c: debit the forgone scrap and the balance to P&L; credit the process figure.

WORKING NOTES

W1 Cost and units

Material 2,000 × ₹5 = 10,000

Direct wages = 900 | Production overheads = 500

Total cost = ₹11,400

Normal loss = 10% of 2,000 = **200 kg**; scrap 200 × ₹3 = **₹600**

Expected output = 1,800 kg | Actual output = 1,850 kg

Abnormal gain = 1,850 – 1,800 = 50 kg

$$\text{Rate} = \frac{11,400 - 600}{2,000 - 200} = \frac{10,800}{1,800} = \text{₹6.00 per kg}$$

Abnormal gain 50 × 6 = **₹300**

Transfer out 1,850 × 6 = **₹11,100**

Process 1 Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Material	2,000	5	10,000	By Normal loss 200 kg × ₹3	200	3	600
To Direct wages	—	—	900	By Transfer to Process 2 W1: 1,850 × ₹6	1,850	6	11,100
To Production overheads	—	—	500				
To Abnormal gain W1: 50 kg × ₹6	50	6	300				
Total	2,050	—	11,700	Total	2,050	—	11,700

Normal Loss Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Process 1 A/c	200	3	600	By Abnormal Gain A/c 50 kg × ₹3 — loss that did not occur	50	3	150
Total	200	—	600	By Cost Ledger Control A/c (scrap sold)	150	3	450
				Total	200	—	600

Abnormal Gain Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Normal Loss A/c scrap income forgone on 50 kg	50	3	150	By Process 1 A/c	50	6	300
To Profit & Loss A/c balance = net gain	—	—	150				
Total	50	—	300	Total	50	—	300

WHY THIS WORKS

The net gain reaching Profit & Loss is only ₹150, not ₹300. The process saved ₹300 of cost by producing 50 extra kg, but it also lost the ₹150 of scrap money those 50 kg would have fetched. Gain is always **cost saved minus scrap forgone**.

FINAL ANSWER

Cost per kg	₹ 6.00
Transferred out (1,850 kg)	₹ 11,100
Abnormal gain at cost (50 kg)	₹ 300
Scrap value forgone	₹ 150
Net abnormal gain to Profit & Loss	₹ 150

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

A product passes through 2 distinct processes A and B. The normal wastage of each process is as follows: Process A – 3% of products entering the process; Process B – 5% of products entering the process.

The wastage of Process A was sold at 50 paise per unit and that of Process B at ₹1 per unit. 10,000 units were issued to Process A at ₹2 per unit. The other expenses were as follows:

Particulars	Process A	Process B
Materials	2,000	3,000
Wages	10,000	16,000
Overheads	2,100	2,375
Actual output (units)	9,500	9,100

Prepare process accounts and other ledger accounts.

READ THE QUESTION FIRST — WHAT TO SPOT

- “3% of products entering the process” — so Process B’s 5% is on the 9,500 units it receives, not on the original 10,000.
- Process A: expected 9,700, actual 9,500 → **abnormal loss 200 units**.
- Process B: expected 9,025, actual 9,100 → **abnormal gain 75 units**.
- Two different scrap rates — 50 paise in A and ₹1 in B. Keep them apart; using one rate for both is a common slip.
- “and other ledger accounts” means: Normal Loss A/c, Abnormal Loss A/c and Abnormal Gain A/c. Draw all three.

THE METHOD FOR THIS TYPE — ALWAYS THESE STEPS

1. Process A: total debits, normal loss 3% of 10,000, expected output, compare, rate, then value the loss and the transfer.
2. Carry A's output *and* A's rate into Process B as the transfer-in.
3. Process B: normal loss 5% of the units received (9,500), then the same sequence.
4. Build the three ledger accounts, remembering the Abnormal Gain → Normal Loss entry.

WORKING NOTES

W1 Process A

Total cost = $(10,000 \times ₹2 = 20,000) + 2,000 + 10,000 + 2,100 = ₹34,100$

Normal loss = 3% of 10,000 = **300 units**; scrap $300 \times ₹0.50 = ₹150$

Expected output 9,700 | Actual 9,500 → **abnormal loss 200 units**

$$\text{Rate} = \frac{34,100 - 150}{10,000 - 300} = \frac{33,950}{9,700} = ₹3.50 \text{ per unit}$$

Abnormal loss $200 \times 3.50 = ₹700$

Transfer to B $9,500 \times 3.50 = ₹33,250$

W2 Process B

Total cost = 33,250 + 3,000 + 16,000 + 2,375 = **₹54,625**

Normal loss = 5% of **9,500** = **475 units**; scrap 475 × ₹1 = **₹475**

Expected output 9,025 | Actual 9,100 → **abnormal gain 75 units**

$$\text{Rate} = \frac{54,625 - 475}{9,500 - 475} = \frac{54,150}{9,025} = \text{₹6.00 per unit}$$

Abnormal gain 75 × 6 = **₹450** (debit side)

Transfer to Finished Stock 9,100 × 6 = **₹54,600**

Process A Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Units introduced	10,000	2	20,000	By Normal loss 300 u × ₹0.50	300	0.50	150
To Materials	—	—	2,000	By Abnormal loss W1: 200 × ₹3.50	200	3.50	700
To Wages	—	—	10,000	By Transfer to Process B W1: 9,500 × ₹3.50	9,500	3.50	33,250
To Overheads	—	—	2,100				
Total	10,000	—	34,100	Total	10,000	—	34,100

Process B Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Transfer from Process A	9,500	3.50	33,250	By Normal loss 475 u × ₹1	475	1	475
To Materials	—	—	3,000	By Transfer to Finished Stock W2: 9,100 × ₹6	9,100	6	54,600
To Wages	—	—	16,000				
To Overheads	—	—	2,375				
To Abnormal gain W2: 75 × ₹6 — DEBIT	75	6	450				
Total	9,575	—	55,075	Total	9,575	—	55,075

Normal Loss Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Process A A/c	300	0.50	150	By Abnormal Gain A/c 75 u × ₹1 (B's scrap rate)	75	1	75
To Process B A/c	475	1	475	By Cost Ledger Control A/c (scrap sold)	700	—	550
Total	775	—	625	Total	775	—	625

Abnormal Loss Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Process A A/c	200	3.50	700	By Cost Ledger Control A/c $200 \times ₹0.50$	200	0.50	100
Total	200	—	700	By Profit & Loss A/c	—	—	600
				Total	200	—	700

Abnormal Gain Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Normal Loss A/c	75	1	75	By Process B A/c	75	6	450
To Profit & Loss A/c	—	—	375				
Total	75	—	450	Total	75	—	450

THE TRAP / WHERE MARKS ARE LOST

- Using ₹0.50 as the scrap rate in Process B. B's wastage sells for ₹1. Two processes, two rates.
- Taking 5% of 10,000 in Process B. It receives only 9,500.
- Netting the abnormal loss of Process A against the abnormal gain of Process B. They are separate accounts and separate P&L entries — never set one off against the other.

FINAL ANSWER

Cost per unit — Process A	₹ 3.50
Cost per unit — Process B	₹ 6.00
Abnormal loss (A, 200 units)	₹ 700 — net to P&L ₹ 600
Abnormal gain (B, 75 units)	₹ 450 — net to P&L ₹ 375
Transferred to Finished Stock (9,100 units)	₹ 54,600

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

The following details are extracted from the costing records of an oil mill for the year ended 31st March, 2015. Purchase of 5,400 tonnes of coconut ₹2,20,000.

Particulars	Crushing	Refining	Finishing
Cost of labour	2,750	1,100	1,650
Electric power	660	396	264
Sundry material	110	2,200	—
Repairs to machinery	308	363	154
Steam	600	495	495
Factory expenses	1,452	726	242

Cost of sacks ₹8,250. 3,200 tonnes of crude oil were produced. 2,600 tonnes of oil produced by the refining process. 2,550 tonnes of refined oil were finished for delivery. Coconut sacks sold ₹440. 1,925 tonnes of coconut residue sold ₹12,100. Loss in weight in crushing 275 tonnes. 500 tonnes of by-products obtained from refining process ₹7,425.

Show the accounts for (a) coconut crushing process, (b) refining process, (c) finishing process, to arrive at the cost per tonne of each process and the total cost per tonne of the finished oil.

READ THE QUESTION FIRST — WHAT TO SPOT

A long question, but it is only Q1 with three deductions. The one thing that makes it easy is the **units reconciliation** — do it first and the whole answer falls out.

- **Crushing:** 5,400 t in. Out: crude oil 3,200 + residue 1,925 + loss in weight 275 = 5,400 . Perfect — no abnormal item.
- **Refining:** 3,200 t in. Out: refined oil 2,600 + by-product 500 = 3,100, so loss in weight = **100 t** (you have to work this out; the question does not give it).
- **Finishing:** 2,600 t in, 2,550 t out, so loss in weight = **50 t**.
- Three items are **credited** to their process because they are saleable: sacks 440 and residue 12,100 in crushing, by-product 7,425 in refining.
- Loss in weight is pure evaporation. It has **no scrap value**, so it carries no rupee figure — only tonnes. Show the tonnes and put a dash in the amount column.

THE METHOD FOR THIS TYPE — ALWAYS THESE STEPS

1. Reconcile tonnes for each stage first. Work out any loss in weight the question has not given you.
2. Debit the process with the material coming in plus all its own expenses.
3. Credit any saleable output that is *not* the main product (sacks, residue, by-product) at its sale value.
4. Credit the loss in weight in tonnes only, amount nil.
5. The balance is the cost of the main output. Divide by its tonnage for cost per tonne.
6. Carry that balance forward as the debit of the next process.

WORKING NOTES

W1 Tonnage reconciliation — do this before anything else

Stage	In (t)	Main output (t)	Other output (t)	Loss in weight (t)
Crushing	5,400	3,200 crude oil	1,925 residue	275 (given)
Refining	3,200	2,600 refined oil	500 by-product	100 (derived)
Finishing	2,600	2,550 finished oil	—	50 (derived)

Refining loss = $3,200 - 2,600 - 500 = \mathbf{100 \text{ tonnes}}$

Finishing loss = $2,600 - 2,550 = \mathbf{50 \text{ tonnes}}$

W2 Cost per tonne at each stage

$$\begin{aligned} \text{Crushing} &= \frac{2,21,590}{3,200 \text{ t}} = \mathbf{₹69.25} \\ \text{Refining} &= \frac{2,19,445}{2,600 \text{ t}} = \mathbf{₹84.40} \\ \text{Finishing} &= \frac{2,22,250}{2,550 \text{ t}} = \mathbf{₹87.16} \end{aligned}$$

(a) Coconut Crushing Process Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Coconut purchased	5,400	—	2,20,000	By Sale of sacks	—	—	440
To Cost of sacks	—	—	8,250	By Sale of coconut residue	1,925	—	12,100
To Cost of labour	—	—	2,750	By Loss in weight evaporation — tonnes only, no value	275	—	—
To Electric power	—	—	660	By Crude oil to Refining W2: 2,34,130 – 440 – 12,100	3,200	69.25	2,21,590
To Sundry material	—	—	110				
To Repairs to machinery	—	—	308				
To Steam	—	—	600				
To Factory expenses	—	—	1,452				
Total	5,400	—	2,34,130	Total	5,400	—	2,34,130

(b) Refining Process Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Crude oil from Crushing	3,200	69.25	2,21,590	By Sale of by-products	500	—	7,425
To Cost of labour	—	—	1,100	By Loss in weight W1: derived as 100 t	100	—	—
To Electric power	—	—	396	By Refined oil to Finishing W2: 2,26,870 – 7,425	2,600	84.40	2,19,445
To Sundry material	—	—	2,200				
To Repairs to machinery	—	—	363				
To Steam	—	—	495				
To Factory expenses	—	—	726				
Total	3,200	—	2,26,870	Total	3,200	—	2,26,870

(c) Finishing Process Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Refined oil from Refining	2,600	84.40	2,19,445	By Loss in weight W1: derived as 50 t	50	—	—
To Cost of labour	—	—	1,650	By Finished oil (cost of delivery) W2: 2,22,250 ÷ 2,550 t	2,550	87.16	2,22,250
To Electric power	—	—	264				
To Repairs to machinery	—	—	154				
To Steam	—	—	495				
To Factory expenses	—	—	242				
Total	2,600	—	2,22,250	Total	2,600	—	2,22,250

THE TRAP / WHERE MARKS ARE LOST

- Treating the sacks, residue and by-product as income in Profit & Loss. They reduce the cost of the process that produced them.
- Giving the loss in weight a rupee value. It evaporated — there is nothing to sell. Tonnes only.
- Not deriving the refining and finishing losses. If you skip them the tonnage columns will not add up and the examiner will see it immediately.
- Dividing the crushing cost by 5,400 tonnes. The cost belongs to the 3,200 tonnes of crude oil that actually came out.

FINAL ANSWER

Cost per tonne — Crushing (crude oil)	₹ 69.25
Cost per tonne — Refining (refined oil)	₹ 84.40
Cost per tonne — Finishing (finished oil)	₹ 87.16
Total cost of 2,550 tonnes delivered	₹ 2,22,250

Closing line: “The total cost per tonne of the finished oil is ₹87.16.”

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

In manufacturing the main product a company processes the incidental waste into two by-products A and B. From the following data relating to the products prepare a comparative profit & loss statement showing individual cost and other details. The total cost up to separation point was ₹3,10,400.

Particulars	Main product	Product A	Product B
Sales	8,00,000	64,000	96,000
Cost after separation	80,000	12,800	14,400
Estimated net profit (% of sales)	?	20%	30%
Estimated selling expenses (% of sales)	20%	10%	15%

READ THE QUESTION FIRST — WHAT TO SPOT

- The main product's profit is a **question mark**. That tells you the whole plan: work out the two by-products completely, take their joint-cost shares away from ₹3,10,400, and whatever is left belongs to the main product.
- You are given **profit % and selling-expense % for A and B**. That is the signal for the **reverse-cost method** — start at sales and climb down.
- All percentages here are **of sales**, not of cost. Read the bracket every time.
- "Comparative profit & loss statement" means one table with a column per product and a total column. Give the total column — it is where the cross-check lives.

THE METHOD FOR THIS TYPE — ALWAYS THESE STEPS

1. For each by-product, start with sales. Subtract profit (% of sales). Subtract selling expenses (% of sales). What remains is that product's **total cost**.
2. Subtract its cost after separation. What remains is its **share of the joint cost**.
3. Add the by-products' shares. Subtract from the total joint cost. The balance is the **main product's share**.
4. Now build the main product's column forwards: joint share + cost after separation + selling expenses = total cost; sales – total cost = profit.
5. Express the main product's profit as a % of its sales — that is the missing '?'.
6. Cross-check: every column's total cost + profit must equal its sales.



The reverse-cost ladder. Every joint / by-product problem in Q10 to Q14 is this one picture.

WORKING NOTES

W1 Product A — climbing down from sales

Sales = 64,000

Less: Net profit 20% of 64,000 = (12,800)
 Less: Selling expenses 10% of 64,000 = (6,400)
Total cost = 44,800
 Less: Cost after separation = (12,800)
Share of joint cost = ₹32,000

W2 Product B — same ladder

Sales = 96,000
 Less: Net profit 30% of 96,000 = (28,800)
 Less: Selling expenses 15% of 96,000 = (14,400)
Total cost = 52,800
 Less: Cost after separation = (14,400)
Share of joint cost = ₹38,400

W3 Main product's share is whatever is left

Total joint cost = 3,10,400
 Less: Product A's share (W1) = (32,000)
 Less: Product B's share (W2) = (38,400)
Main product's share of joint cost = ₹2,40,000

Comparative Profit and Loss Statement

Particulars	Main product	Product A	Product B	Total
Sales	8,00,000	64,000	96,000	9,60,000
Share of joint cost (W1-W3)	2,40,000	32,000	38,400	3,10,400
Cost after separation	80,000	12,800	14,400	1,07,200
Selling expenses	1,60,000	6,400	14,400	1,80,800
Total cost	4,80,000	51,200	67,200	5,98,400
Net profit (Sales – Total cost)	3,20,000	12,800	28,800	3,61,600
Net profit as % of sales	40%	20%	30%	37.67%

Selling expenses: main 20% of 8,00,000 = 1,60,000 | A 10% of 64,000 = 6,400 | B 15% of 96,000 = 14,400.

THE TRAP / WHERE MARKS ARE LOST

- Taking the profit % on **cost** instead of on sales. Here everything is on sales.
- Forgetting to subtract selling expenses on the way down. If you skip them, A's joint share comes to 38,400 instead of 32,000 and the whole answer shifts.
- Apportioning the joint cost by sales value or by weight. The question gives you profit and selling-expense percentages precisely so you use the reverse-cost method.

FINAL ANSWER

Joint cost — Main product	₹ 2,40,000
Joint cost — Product A	₹ 32,000
Joint cost — Product B	₹ 38,400
Main product's net profit	₹ 3,20,000 (40% of sales)
Total profit of all three	₹ 3,61,600

Q11 Selling expenses have to be derived

Joint & by-product · p.11

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

A factory producing an article 'X' also produces product 'Y' which is further processed into a finished product. The joint cost of manufacturing is: Materials 50,000; Labour 30,000; Overheads 20,000 — total ₹1,00,000. The subsequent costs are as follows:

Particulars	X	Y
Materials	30,000	15,000
Labour	14,000	10,000
Overheads	6,000	5,000
Total	50,000	30,000
Selling price	1,60,000	80,000
Estimated profit on selling price	25%	20%

Assume that the selling and distribution expenses are in proportion to the selling price. Show how you would apportion the joint cost of manufacturing and prepare a statement showing the cost of production of both products and the ledger accounts.

READ THE QUESTION FIRST — WHAT TO SPOT

- This one has a hidden step. The **selling and distribution expenses are not given** — you must derive the total, then split it.
- How to derive it: total sales 2,40,000 minus total profit gives total cost. Take away the costs you already know (joint 1,00,000 + subsequent 80,000) and the gap must be the S&D expenses.
- "In proportion to the selling price" — so split in the ratio $1,60,000 : 80,000 = 2 : 1$.
- Then it is the ordinary reverse-cost ladder for each product.

THE METHOD FOR THIS TYPE — ALWAYS THESE STEPS

1. Compute each product's profit from the given % of selling price. Add them.
2. Total cost = Total sales — Total profit.
3. **S&D expenses = Total cost — joint cost — subsequent costs.** This is the derived figure.
4. Split the S&D expenses in the ratio of selling prices.
5. For each product climb down: Sales — profit = total cost; then — S&D — subsequent cost = share of joint cost.
6. Check the two joint shares add back to 1,00,000.

WORKING NOTES**W1 Deriving the selling & distribution expenses**

Profit on X = 25% of 1,60,000 = 40,000

Profit on Y = 20% of 80,000 = 16,000

Total profit = 56,000

Total sales = 1,60,000 + 80,000 = 2,40,000

Total cost = 2,40,000 – 56,000 = **1,84,000**
 Known costs = joint 1,00,000 + subsequent 80,000 = 1,80,000
 $\therefore$ **S&D expenses** = **1,84,000 – 1,80,000 = ₹4,000**

Split 2 : 1 $\rightarrow$ X = 4,000 $\times$ 2/3 = **2,666.67** | Y = 4,000 $\times$ 1/3 = **1,333.33**

W2 Apportionment of the joint cost

Particulars	Product X	Product Y	Total
Selling price	1,60,000	80,000	2,40,000
Less: Estimated profit	(40,000)	(16,000)	(56,000)
Total cost	1,20,000	64,000	1,84,000
Less: Selling & distribution (W1)	(2,666.67)	(1,333.33)	(4,000)
Less: Subsequent cost	(50,000)	(30,000)	(80,000)
Share of joint cost	67,333.33	32,666.67	1,00,000

Cross-check: 67,333.33 + 32,666.67 = 1,00,000 exactly the joint cost given. That agreement is your proof the method is right.

Statement of Cost of Production and Profit

Particulars	Product X	Product Y	Total
Share of joint cost (W2)	67,333.33	32,666.67	1,00,000
Subsequent cost — materials	30,000	15,000	45,000
Subsequent cost — labour	14,000	10,000	24,000
Subsequent cost — overheads	6,000	5,000	11,000
Cost of production	1,17,333.33	62,666.67	1,80,000
Selling & distribution expenses	2,666.67	1,333.33	4,000
Total cost	1,20,000	64,000	1,84,000
Sales	1,60,000	80,000	2,40,000
Profit	40,000	16,000	56,000
Profit as % of selling price	25%	20%	23.33%

Product X Account

Particulars	Amount ₹	Particulars	Amount ₹
To Joint cost apportioned	—	By Sales	—
To Materials	—		
To Labour	—		
To Overheads	—		
To Selling & distribution W1: 4,000 $\times$ 2/3	—		
To Profit & Loss A/c (profit)	—		
Total	—	Total	—

Product Y Account

Particulars	Amount ₹	Particulars	Amount ₹
To Joint cost apportioned	—	By Sales	—
To Materials	—		
To Labour	—		
To Overheads	—		
To Selling & distribution W1: $4,000 \times 1/3$	—		
To Profit & Loss A/c (profit)	—		
Total	—	Total	—

WHY THIS WORKS

The reason the S&D expenses can be derived at all is that the question fixes the profit on *both* products. Once profit is fixed, total cost is fixed, and any cost you have not been told about must be the difference. Whenever a joint-cost question gives you every profit percentage but leaves one cost unnamed, this is the trick being tested.

FINAL ANSWER

Joint cost apportioned to X	₹ 67,333.33
Joint cost apportioned to Y	₹ 32,666.67
Selling & distribution expenses (derived)	₹ 4,000
Cost of production — X	₹ 1,17,333.33
Cost of production — Y	₹ 62,666.67
Total profit	₹ 56,000

Q12 When the printed figures do not reconcile

Joint & by-product · p.11-12

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

In a manufacturing concern a certain product is manufactured which also yields 2 by-products. The joint expenses of manufacture are: Materials 8,500; Labour 9,000; Overheads 7,500 — total ₹25,000. Subsequent expenses are:

Particulars	A	B	C
Materials	2,500	1,200	1,400
Labour	1,900	1,600	2,000
Overheads	1,500	900	1,050
Total	5,900	3,700	4,450

The selling prices are A — 30,000; B — 20,000 and C — 15,000. Estimated profits on selling prices are A — 40%; B — 30% and C — 25%. Show the apportionment of joint cost and prepare the product accounts.

READ THE QUESTION FIRST — WHAT TO SPOT

- Same ladder as Q10 and Q11, but with **no selling expenses** mentioned — so the ladder is shorter: Sales – profit – subsequent cost = share of joint cost.
- **Read the warning below before you write your answer.** The figures in this question do not reconcile, and knowing how to handle that is worth more than getting a tidy number.

THE METHOD FOR THIS TYPE — ALWAYS THESE STEPS

1. For each product: Sales – profit (% of sales) = total cost.
2. Total cost – subsequent cost = **notional share of joint cost**.
3. Add the three notional shares and **compare with the actual joint cost**.
4. If they agree, you are finished. **If they do not agree** (as here), apportion the *actual* joint cost in the *ratio* of the notional shares.
5. Recompute each product's real profit using its apportioned share, and state that the actual profit differs from the estimate.

WORKING NOTES

W1 Notional share of joint cost by reverse cost

Particulars	A	B	C	Total
Selling price	30,000	20,000	15,000	65,000
Less: Estimated profit (40% / 30% / 25%)	(12,000)	(6,000)	(3,750)	(21,750)
Total cost	18,000	14,000	11,250	43,250
Less: Subsequent expenses	(5,900)	(3,700)	(4,450)	(14,050)
Notional share of joint cost	12,100	10,300	6,800	29,200

W2 The figures do not reconcile — and here is how to deal with it

Notional shares add to = **29,200**

Actual joint expenses are only = **25,000**

Difference = **₹4,200** — the estimated profits are more optimistic than the actual cost structure allows.

The joint cost that actually exists is 25,000, so that is all you can apportion. Apportion it in the **ratio of the notional shares** 12,100 : 10,300 : 6,800.

Product	Notional share	Working	Apportioned joint cost
A	12,100	$25,000 \times 12,100 / 29,200$	10,359.59
B	10,300	$25,000 \times 10,300 / 29,200$	8,818.49
C	6,800	$25,000 \times 6,800 / 29,200$	5,821.92
Total	29,200		25,000.00

Statement of Cost and Actual Profit

Particulars	A	B	C	Total
Apportioned joint cost (W2)	10,359.59	8,818.49	5,821.92	25,000
Subsequent expenses	5,900	3,700	4,450	14,050

Particulars	A	B	C	Total
Total cost	16,259.59	12,518.49	10,271.92	39,050
Sales	30,000	20,000	15,000	65,000
Actual profit	13,740.41	7,481.51	4,728.08	25,950
Actual profit as % of sales	45.80%	37.41%	31.52%	39.92%
Estimated profit % (from the question)	40%	30%	25%	—

Product A Account

Particulars	Amount ₹	Particulars	Amount ₹
To Joint cost apportioned W2	—	By Sales	—
To Subsequent expenses	—		
To Profit & Loss A/c (profit)	—		
Total	—	Total	—

Product B Account

Particulars	Amount ₹	Particulars	Amount ₹
To Joint cost apportioned W2	—	By Sales	—
To Subsequent expenses	—		
To Profit & Loss A/c (profit)	—		
Total	—	Total	—

Product C Account

Particulars	Amount ₹	Particulars	Amount ₹
To Joint cost apportioned W2	—	By Sales	—
To Subsequent expenses	—		
To Profit & Loss A/c (profit)	—		
Total	—	Total	—

THE TRAP / WHERE MARKS ARE LOST

Please read this one carefully — it is about your exam, not just this sum.

- The notional shares total 29,200 but the joint expenses are 25,000. As printed, the question cannot give a tidy answer. This happens in workbooks.
- The **wrong** response is to write 12,100 / 10,300 / 6,800 as your apportionment. Those figures add to 29,200, which is ₹4,200 more joint cost than the company actually incurred.
- The **right** response is what W2 does: apportion the real 25,000 in the ratio of the notional shares, then show the actual profit percentages and note that they exceed the estimates. Write one sentence saying so. An examiner rewards a student who spots the inconsistency and handles it openly.
- **Do check this one with your professor** — if he teaches the plain notional-share answer, give that instead. Either way, understand why the two differ.

FINAL ANSWER

Apportioned joint cost — A	₹ 10,359.59
Apportioned joint cost — B	₹ 8,818.49
Apportioned joint cost — C	₹ 5,821.92
Total actual profit	₹ 25,950
Notional shares (for the ratio only)	12,100 : 10,300 : 6,800

Q13 Three joint products, full ladder

Joint & by-product · p.12

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

Bright Chemical Ltd. electrolyses common salt to obtain 3 joint products — caustic soda, chlorine and hydrogen. During a costing period, the expenditure relating to the inputs of the common process amounted to ₹3,50,000. After separation, expenses amounting to ₹1,60,000, ₹75,000 and ₹10,000 were incurred for caustic soda, chlorine and hydrogen respectively. The entire production was sold and ₹3,75,000, ₹2,50,000 and ₹60,000 were realised for caustic soda, chlorine and hydrogen respectively. The selling expenses were estimated at 5% of realisation from sale. The profit is 15%, 10% and 5% of realisation from sale of caustic soda, chlorine and hydrogen respectively.

Draw a columnar statement showing the apportionment of joint cost and the profitability of each product.

READ THE QUESTION FIRST — WHAT TO SPOT

- The full four-rung ladder this time: Sales — profit — selling expenses — separation expenses = share of joint cost.
- Selling expenses are **5% of sales for all three** — one rate, three amounts.
- These are **joint products**, not by-products. That changes nothing about the arithmetic; it only means all three are commercially important, so none is credited to another.
- Like Q12, the derived shares here **do not** add up to the joint cost given. The same ratio treatment applies. Do not be alarmed — be systematic.

THE METHOD FOR THIS TYPE — ALWAYS THESE STEPS

1. Build the ladder in a single columnar table — one column per product, one total column.
2. Row 1 Sales. Row 2 less profit. Row 3 less selling expenses (5% of sales). Row 4 gives total cost.
3. Row 5 less separation expenses. Row 6 gives the notional joint-cost share.
4. Total row 6 and compare with the actual joint cost of 3,50,000.
5. Apportion the actual 3,50,000 in the ratio of the notional shares.
6. Rebuild each column forwards to show the actual profitability, and comment on how it compares with the estimate.

WORKING NOTES

W1 The reverse-cost ladder, all three products together

Particulars	Caustic soda	Chlorine	Hydrogen	Total
Sales realisation	3,75,000	2,50,000	60,000	6,85,000
Less: Profit (15% / 10% / 5% of sales)	(56,250)	(25,000)	(3,000)	(84,250)
Less: Selling expenses (5% of sales)	(18,750)	(12,500)	(3,000)	(34,250)
Total cost	3,00,000	2,12,500	54,000	5,66,500

Particulars	Caustic soda	Chlorine	Hydrogen	Total
Less: Expenses after separation	(1,60,000)	(75,000)	(10,000)	(2,45,000)
Notional share of joint cost	1,40,000	1,37,500	44,000	3,21,500

W2 Reconciling with the actual joint cost

Notional shares total = **3,21,500**

Actual joint expenditure = **3,50,000**

Shortfall = **₹28,500** — the joint process cost more than the estimated profits allow for, so the actual profits will be *lower* than estimated.

Product	Notional share	Working	Apportioned joint cost
Caustic soda	1,40,000	$3,50,000 \times 1,40,000 / 3,21,500$	1,52,410.58
Chlorine	1,37,500	$3,50,000 \times 1,37,500 / 3,21,500$	1,49,688.96
Hydrogen	44,000	$3,50,000 \times 44,000 / 3,21,500$	47,900.46
Total	3,21,500		3,50,000.00

Columnar Statement of Apportionment and Profitability

Particulars	Caustic soda	Chlorine	Hydrogen	Total
Apportioned joint cost (W2)	1,52,410.58	1,49,688.96	47,900.46	3,50,000
Expenses after separation	1,60,000	75,000	10,000	2,45,000
Selling expenses (5% of sales)	18,750	12,500	3,000	34,250
Total cost	3,31,160.58	2,37,188.96	60,900.46	6,29,250
Sales realisation	3,75,000	2,50,000	60,000	6,85,000
Actual profit / (loss)	43,839.42	12,811.04	(900.46)	55,750
Actual profit as % of sales	11.69%	5.12%	(1.50%)	8.14%
Estimated profit % (question)	15%	10%	5%	—

WHY THIS WORKS

Look at the hydrogen column: an actual **loss** of ₹900.46 against an estimated 5% profit. That is a real finding, not an error — the joint process cost ₹28,500 more than the estimates assumed, and hydrogen, having the smallest sales, cannot carry its share. Write that sentence in your answer. It shows you can read a statement instead of only filling it in.

FINAL ANSWER

Apportioned joint cost — Caustic soda **₹ 1,52,410.58**

Apportioned joint cost — Chlorine **₹ 1,49,688.96**

Apportioned joint cost — Hydrogen **₹ 47,900.46**

Total actual profit **₹ 55,750**

Notional shares (ratio basis) **1,40,000 : 1,37,500 : 44,000**

State clearly that the notional shares totalled ₹3,21,500 against actual joint cost of ₹3,50,000, and that you apportioned the actual figure in that ratio.

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

A by-product 'Beta' is derived in the course of manufacturing product 'Alpha'. The by-product is further processed for sale. From the following data available from the records, prepare an account showing the cost per kg of the product 'Alpha' and the by-product 'Beta'.

Particulars	Joint expenses	Separate — Alpha	Separate — Beta
Materials	10,000	6,000	500
Labour	7,000	5,000	2,000
Overheads	2,500	1,500	600
Total	19,500	12,500	3,100

The quantities produced during the period under consideration were: Alpha — 1,000 kgs and Beta — 500 kgs. The selling price was ₹120/kg on which the profit earned was 30%.

READ THE QUESTION FIRST — WHAT TO SPOT

- A **by-product** problem: Alpha is the main product, Beta is incidental. The normal treatment is to work out Beta's share of the joint cost by reverse cost and leave the balance for Alpha.
- Total cost available to share out = joint 19,500 + Alpha's own 12,500 + Beta's own 3,100 = **₹35,100**. Keep that number in your head — it is the ceiling on every cost figure in this problem.
- **Now look at the selling price.** 500 kg of Beta at 120 would be 60,000 of sales, and 1,000 kg of Alpha at 120 would be 1,20,000. Either one is far bigger than the total cost of 35,100. So the reverse-cost ladder is going to break. Read W2 before you attempt this in the exam.

THE METHOD FOR THIS TYPE — ALWAYS THESE STEPS

1. Identify the main product and the by-product.
2. Apply the reverse-cost ladder to the **by-product**: Sales – profit – its own separate expenses = its share of the joint cost.
3. Joint cost less the by-product's share = the main product's share.
4. Main product cost = its joint share + its own separate expenses. Divide by its kg for cost per kg. Same for the by-product.
5. **Always sanity-check** that no share is negative and that the shares add back to the joint cost. If they do not, the data is faulty — see W2.

WORKING NOTES

W1 The reverse-cost attempt (taking ₹120/kg as Beta's price)

Beta sales = 500 kg × 120 = 60,000

Less: Profit 30% of 60,000 = (18,000)

Total cost of Beta = 42,000

Less: Beta's separate expenses = (3,100)

Beta's share of joint cost = **38,900**

But the whole joint cost is only **19,500** → Alpha's share would be **19,500 – 38,900 = (₹19,400)**, a negative cost.

W2 THE DATA AS PRINTED CANNOT BE SOLVED — AND WHAT TO DO ABOUT IT

A cost can never be negative. Whichever product you assign the ₹120/kg to, the sales value dwarfs the total cost of ₹35,100, so the ladder produces an impossible share. Check the figures against the original in your workbook — most likely the selling price, the profit percentage or the joint expenses has a typographical error.

In the exam, do this: write one line stating the inconsistency, then solve the problem on the next most defensible basis and say which basis you used. You will be given credit for the method. The standard fallback for a by-product whose selling price is unusable is to apportion the joint cost on **physical quantity**, which is shown in W3.

W3 Workable solution — joint cost apportioned on physical quantity

Quantities Alpha 1,000 kg : Beta 500 kg = **2 : 1**

Alpha's share = $19,500 \times 2/3 = \text{₹}13,000$

Beta's share = $19,500 \times 1/3 = \text{₹}6,500$

Particulars	Alpha	Beta	Total
Share of joint cost (2 : 1)	13,000	6,500	19,500
Add: Separate expenses	12,500	3,100	15,600
Total cost	25,500	9,600	35,100
Quantity produced (kg)	1,000	500	1,500
Cost per kg	25.50	19.20	—

Alpha Account (main product)

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Share of joint cost W3: $19,500 \times 2/3$	1,000	13	13,000	By Cost of production transferred $25,500 \div 1,000 \text{ kg}$	1,000	25.50	25,500
To Separate materials	—	—	6,000				
To Separate labour	—	—	5,000				
To Separate overheads	—	—	1,500				
Total	1,000	—	25,500	Total	1,000	—	25,500

Beta Account (by-product)

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Share of joint cost W3: $19,500 \times 1/3$	500	13	6,500	By Cost of production transferred $9,600 \div 500 \text{ kg}$	500	19.20	9,600
To Separate materials	—	—	500				
To Separate labour	—	—	2,000				
To Separate overheads	—	—	600				
Total	500	—	9,600	Total	500	—	9,600

FINAL ANSWER

Alpha — total cost	₹ 25,500
Alpha — cost per kg	₹ 25.50
Beta — total cost	₹ 9,600

Beta — cost per kg

₹ 19.20

Basis used

Physical quantity 2 : 1 (see W2 for why)

Ask your professor about this one. If he wants the reverse-cost method, the corrected selling price is needed. Show W1, W2 and W3 exactly as laid out above and you will not lose method marks either way.

Q15 Profit as % of cost, plus unrealised profit

Inter-process profit · p.12-13

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

A product passes through two processes X and Y. The output of Process X is charged to Process Y at a price which includes profit of **20% on actual cost** and the output of Process Y is charged to Finished Stock A/c at a price which includes **10% profit on actual cost**. The following data is provided for the month of July.

Particulars	X	Y
Material (2,500 units)	1,250	—
Labour	625	500
Overheads	1,875	750
Indirect material	—	1,250

There was no partly finished WIP. Out of the finished stock, 1,500 units had been sold for ₹7,500. Prepare the Process A/c and Finished Stock A/c.

READ THE QUESTION FIRST — WHAT TO SPOT

- This is the **inter-process profit** topic. Each process sells to the next at a mark-up, so the transfer price is bigger than the true cost.
- **“20% on actual cost”** — profit is 20% of cost, so if cost is 3,750 the profit is 750. (Contrast Q16, where profit is a % of the transfer price — a different calculation.)
- There is **no WIP inside the processes**, which makes this the easy version. The only stock is in Finished Stock: 2,500 units produced, 1,500 sold, so **1,000 units left**.
- That closing stock of 1,000 units carries profit added by X and by Y which the company has not actually earned yet, because the goods are still on the shelf. That is the **unrealised profit**, and a reserve must be created for it.

THE METHOD FOR THIS TYPE — ALWAYS THESE STEPS

1. Total Process X's own costs. That is its **actual cost**.
2. Add the stated profit percentage **of that cost**. The sum is the transfer price to Y.
3. Debit Process Y with the transfer price (not the cost), add Y's own costs, total to get Y's actual cost.
4. Add Y's profit percentage of that total. The sum transfers to Finished Stock.
5. In Finished Stock, work out the per-unit transfer value, split it between units sold and units in hand, and compute the profit on sale.
6. **Unrealised profit** = total profit loaded by the processes $\times \frac{\text{closing stock units}}{\text{total units produced}}$. Create the reserve for it.

WORKING NOTES

W1 Process X

Material 1,250 + Labour 625 + Overheads 1,875

Actual cost = ₹3,750

Profit 20% of 3,750 = **₹750**

Transfer price to Process Y = ₹4,500

W2 Process Y

Transfer from X 4,500 + Indirect material 1,250 + Labour 500 + Overheads 750

Actual cost = ₹7,000

Profit 10% of 7,000 = **₹700**

Transfer to Finished Stock = ₹7,700 for 2,500 units

W3 Finished Stock

Value per unit = $7,700 \div 2,500 = \text{₹}3.08$

Cost of 1,500 units sold = $1,500 \times 3.08 = \text{₹}4,620$

Closing stock 1,000 units = $1,000 \times 3.08 = \text{₹}3,080$

Profit on sale = $7,500 - 4,620 = \text{₹}2,880$

W4 Reserve for unrealised profit — the marks everyone drops

Profit loaded by the processes = $750 (X) + 700 (Y) = \text{₹}1,450$

True cost of the 2,500 units = $7,700 - 1,450 = 6,250$ (check: $3,750 + 1,250 + 500 + 750 = 6,250$ ✓)

Closing stock is 1,000 of 2,500 units = **40%**

$$\text{Reserve} = \frac{1,000 \text{ units}}{2,500 \text{ units}} \times \text{₹}1,450 = \text{₹}580$$

Equivalently $3,080 \times 1,450 / 7,700 = 580$. Both routes give the same figure.

Process X Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Material	2,500	—	1,250	By Transfer to Process Y W1: cost 3,750 + profit 750	2,500	1.80	4,500
To Labour	—	—	625				
To Overheads	—	—	1,875				
Actual cost	—	—	3,750				
To Profit — 20% on cost W1: 20% of 3,750	—	—	750				
Total	2,500	—	4,500	Total	2,500	—	4,500

Process Y Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Transfer from Process X	2,500	1.80	4,500	By Transfer to Finished Stock W2: cost 7,000 + profit 700	2,500	3.08	7,700
To Indirect material	—	—	1,250				
To Labour	—	—	500				
To Overheads	—	—	750				
Actual cost	—	—	7,000				
To Profit — 10% on cost W2: 10% of 7,000	—	—	700				
Total	2,500	—	7,700	Total	2,500	—	7,700

Finished Stock Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Transfer from Process Y	2,500	3.08	7,700	By Sales	1,500	5	7,500
To Profit & Loss A/c (profit on sale) W3: 7,500 – 4,620	—	—	2,880	By Closing stock W3: 1,000 × ₹3.08	1,000	3.08	3,080
Total	2,500	—	10,580	Total	2,500	—	10,580

THE TRAP / WHERE MARKS ARE LOST

- **Confusing “20% on cost” with “20% on transfer price”.** On cost: profit = $0.20 \times \text{cost}$. On transfer price: profit = $0.20 \times \text{TP}$, which means profit = $0.25 \times \text{cost}$. Q16 uses the second form — compare the two side by side once and you will never mix them again.
- Forgetting the reserve for unrealised profit entirely. It is a separate, named figure and questions ask for it explicitly.
- Charging Process Y with X's cost of 3,750. Y is charged the *transfer price* of 4,500 — that is the whole point of the topic.

FINAL ANSWER

Process X — actual cost / transfer price	₹ 3,750 / ₹ 4,500
Process Y — actual cost / transfer price	₹ 7,000 / ₹ 7,700
Profit on sale of 1,500 units	₹ 2,880
Closing stock (1,000 units)	₹ 3,080
Reserve for unrealised profit	₹ 580
Total profit shown (750 + 700 + 2,880)	₹ 4,330
Real profit after the reserve	₹ 3,750

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

A product passes through three processes to completion, known as A, B and C. The output of each process is charged to the next process at a price calculated to give a profit of **20% on the transfer price**. The output of Process C is charged to finished stock on a similar basis. There was no partly finished WIP in any process on December 31st, on which day the following information was obtained.

Particulars	Process A	Process B	Process C
Materials	4,000	6,000	2,000
Labour	6,000	4,000	8,000
Stock: 31st Dec	2,000	4,000	6,000

There was no stock in hand on Jan 1st and overheads were ignored. Of the goods passed into finished stock, ₹4,000 remained in hand on Dec 31st and the balance has been sold for ₹36,000. Show the Process A/c and calculate reserve for unrealised profits.

READ THE QUESTION FIRST — WHAT TO SPOT

- **“20% on the transfer price”** — not on cost. So if the transfer price is 10,000 the profit is 2,000 and the cost is 8,000. Working backwards: Transfer price = Cost ÷ 0.80.
- **Every process has closing stock this time.** That stock is deducted *before* you load the profit, because you only charge profit on what actually leaves.
- Because stock sits in B and C, and those stocks contain profit passed on from earlier processes, you must track **cost and profit in separate columns**. This is the **three-column format** — Cost | Profit | Total. Use it and this problem is straightforward; try to do it in one column and you will not be able to find the reserve.
- Process A's own stock contains **no** profit — nothing has been marked up into it yet. Only B, C and Finished Stock hold unrealised profit.

THE METHOD FOR THIS TYPE — ALWAYS THESE STEPS

1. Draw three columns for every process: **Cost, Profit, Total**.
2. Enter the transfer-in split into its cost and profit parts, then the process's own costs (all in the Cost column — own costs carry no profit).
3. Total the three columns.
4. Deduct closing stock. Split it in the **same cost : profit ratio as the total debit**: stock cost part = stock $\times \frac{\text{total cost}}{\text{total debit}}$.
5. What remains is the cost of goods transferred. Transfer price = that cost ÷ 0.80; the difference is the profit for this process.
6. Repeat for the next process. In Finished Stock do the same split for the goods in hand.
7. **Reserve = the Profit column of every closing stock added together.**
8. Cross-check: total apparent profit – reserve = the realised profit from Finished Stock.

WORKING NOTES

W1 Turning “20% on transfer price” into arithmetic

$$\text{Transfer price} = \frac{\text{Cost of goods transferred}}{0.80} \quad \text{and} \quad \text{Profit} = 20\% \text{ of that price}$$

Because Cost = 80% of the transfer price. Note this is the same as 25% on cost — but always work from the price, as the question states it.

W2 Splitting each closing stock into cost and profit

Process	Total debit	of which Cost	Closing stock	Stock — cost part	Stock — profit part
A	10,000	10,000	2,000	2,000	Nil
B	20,000	18,000	4,000	$4,000 \times 18/20 = 3,600$	400
C	30,000	24,400	6,000	$6,000 \times 24,400/30,000 = 4,880$	1,120
Finished Stock	30,000	19,520	4,000	$4,000 \times 19,520/30,000 = 2,602.67$	1,397.33
Total reserve					2,917.33

Process A Account

Particulars	Cost ₹	Profit ₹	Total ₹
To Materials	4,000	—	4,000
To Labour	6,000	—	6,000
Total debit	10,000	—	10,000
Less: Closing stock	(2,000)	—	(2,000)
Cost of goods transferred	8,000	—	8,000
Add: Profit — 20% on transfer price $8,000 \div 0.80 = 10,000$; profit 2,000	—	2,000	2,000
Transferred to Process B	8,000	2,000	10,000

Process B Account

Particulars	Cost ₹	Profit ₹	Total ₹
To Transfer from Process A	8,000	2,000	10,000
To Materials	6,000	—	6,000
To Labour	4,000	—	4,000
Total debit	18,000	2,000	20,000
Less: Closing stock W2: 4,000 split 18/20	(3,600)	(400)	(4,000)
Cost of goods transferred	14,400	1,600	16,000
Add: Profit $16,000 \div 0.80 = 20,000$; profit 4,000	—	4,000	4,000
Transferred to Process C	14,400	5,600	20,000

Process C Account

Particulars	Cost ₹	Profit ₹	Total ₹
To Transfer from Process B	14,400	5,600	20,000
To Materials	2,000	—	2,000
To Labour	8,000	—	8,000

Particulars	Cost ₹	Profit ₹	Total ₹
Total debit	24,400	5,600	30,000
Less: Closing stock W2: 6,000 split 24,400/30,000	(4,880)	(1,120)	(6,000)
Cost of goods transferred	19,520	4,480	24,000
Add: Profit $24,000 \div 0.80 = 30,000$; profit 6,000	—	6,000	6,000
Transferred to Finished Stock	19,520	10,480	30,000

Finished Stock Account

Particulars	Cost ₹	Profit ₹	Total ₹
To Transfer from Process C	19,520	10,480	30,000
Less: Closing stock in hand W2: 4,000 split 19,520/30,000	(2,602.67)	(1,397.33)	(4,000)
Cost of sales	16,917.33	9,082.67	26,000
Sales	—	—	36,000
Profit on sale $36,000 - 26,000$	—	10,000	10,000
Total profit realised	—	19,082.67	—

Reconciliation — proof that the reserve is right

Particulars	Amount ₹
Profit shown by Process A	2,000
Profit shown by Process B	4,000
Profit shown by Process C	6,000
Profit on sale in Finished Stock	10,000
Total apparent profit	22,000.00
Less: Reserve for unrealised profit (W2)	(2,917.33)
Actual (realised) profit	19,082.67
Agrees with the Finished Stock A/c profit column ✓	19,082.67

WHY THIS WORKS

The reserve exists because the company has already recorded ₹22,000 of profit, but ₹2,917.33 of that is sitting inside goods it still owns. You cannot count profit on goods you have not sold. Removing the reserve brings the books back to the profit actually earned — and the fact that it reconciles to the paisa is your proof the whole answer is right.

FINAL ANSWER

Process A — profit / transfer price	₹ 2,000 / ₹ 10,000
Process B — profit / transfer price	₹ 4,000 / ₹ 20,000
Process C — profit / transfer price	₹ 6,000 / ₹ 30,000
Profit on sale	₹ 10,000
Total apparent profit	₹ 22,000
Reserve for unrealised profit	₹ 2,917.33
Actual realised profit	₹ 19,082.67

Reserve built up as: A nil + B 400 + C 1,120 + Finished Stock 1,397.33 = ₹2,917.33.

Equivalent Production

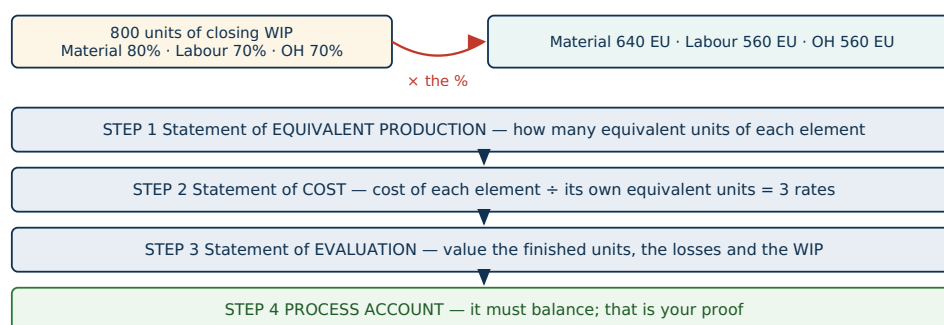
Problems 17 to 24 · workbook pages 13 to 16

THE SINGLE IDEA BEHIND ALL EIGHT PROBLEMS

Until now every unit was either finished or lost. Now some units are **half-finished** at the end of the month. You cannot divide cost by “3,000 finished units + 800 half-finished units”, because those 800 have not absorbed a full unit’s worth of cost.

So you convert them. **800 units that are 70% complete are treated as 560 complete units.** That converted figure is called an **equivalent unit**, and once you have it the problem collapses back into the ordinary “total cost ÷ units” you already know.

The one extra wrinkle: material, labour and overhead are usually at *different* stages of completion. So you do the conversion **three times** and get **three separate rates**.



Four statements, always in this order, in every one of Q17 to Q24. Never skip one.

The rules for the equivalent-production statement

What appears in the output column	Material	Labour	Overhead	Why
Units completed & transferred	100%	100%	100%	They are finished. Full cost in every element.
Normal loss	Nil	Nil	Nil	This is the rule students forget. Normal loss gets zero equivalent units — its cost is absorbed by the good units, so giving it units would double-count.
Abnormal loss	its % of completion	its %	its %	It is a real loss of real cost, so it must carry equivalent units. If the question says the scrapped units were 100% complete, use 100%.
Abnormal gain	100%	100%	100%	Always 100%, and it is deducted in the statement, not added.
Closing WIP	its stated %		its stated %	The whole reason the topic exists.

What appears in the output column	Material	Labour	Overhead	Why
		its stated %		
Opening WIP — FIFO only	the % still to do	the % to do	the % to do	Under FIFO the opening WIP was partly done last month, so only the balance belongs to this month. 40% complete means 60% to do.

FIFO VERSUS AVERAGE — THE ONLY DIFFERENCE THAT MATTERS

	FIFO method	Average method
Opening WIP units in the statement	Only the unfinished portion (the % still to do)	Not shown separately — the opening units are inside the completed figure at 100%
Opening WIP cost	Kept apart. Added back only at the evaluation stage, on top of the cost of finishing it.	Merged with this month's cost before dividing.
Which costs go into the rate	This month's costs only	Opening WIP cost + this month's costs
Use it when	The question says FIFO, or gives the opening WIP's degree of completion element by element	The question says average, or gives the opening WIP cost as one lump you cannot split

Q21 and Q22 are FIFO. Q23 is average. **Q24 asks for both** on the same data, which is the best possible way to see the difference — work it and compare the two totals.

Q17 The four statements, cleanest case

Equivalent production · p.13

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

In Process A on March 1st, there was no opening work-in-progress. During the month of March, 2,000 units of material was issued at a cost of ₹18,000. Labour and overheads totalled to ₹9,000 and ₹6,600 respectively. On 31st March 1,500 units were completed and transferred to the next process. Of the remaining 500 units which are incomplete, degree of completion was: Material 100%, Labour 60% and Overheads 30%.

Prepare (a) Statement of equivalent production (b) Statement of cost (c) Statement of evaluation (d) Process account.

READ THE QUESTION FIRST — WHAT TO SPOT

- **No opening WIP and no losses.** The simplest possible version — the perfect one to learn the four statements on.
- Units: 2,000 in = 1,500 transferred + 500 closing WIP. Balanced, so nothing is lost.
- **Material is 100% complete.** That is normal — material is usually issued in full at the start, so closing WIP has all its material but only part of its labour and overhead.
- Three different percentages means **three different rates**. Do not average them into one.

(a) Statement of Equivalent Production

Output / Particulars	Units	Material %	Material EU	Labour %	Labour EU	Overhead %	Overhead EU
Units completed & transferred	1,500	100	1,500	100	1,500	100	1,500
Closing work-in-progress	500	100	500	60	300	30	150
Total equivalent units			2,000		1,800		1,650

(b) Statement of Cost

Element	Cost ₹	Equivalent units	Cost per unit ₹
Material	18,000	2,000	9.00
Labour	9,000	1,800	5.00
Overhead	6,600	1,650	4.00
Total cost per unit			18.00

(c) Statement of Evaluation

Particulars	Working	Amount ₹
Units completed & transferred	1,500 units × ₹18.00	27,000
Closing work-in-progress		
Material	500 EU × ₹9.00	4,500
Labour	300 EU × ₹5.00	1,500
Overhead	150 EU × ₹4.00	600
Value of closing WIP		6,600
Total		33,600

(d) Process A Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Material	2,000	—	18,000	By Transfer to next process 1,500 × ₹18	1,500	18	27,000
To Labour	—	—	9,000	By Closing WIP statement (c)	500	—	6,600
To Overheads	—	—	6,600				
Total	2,000	—	33,600	Total	2,000	—	33,600

THE TRAP / WHERE MARKS ARE LOST

- Dividing all three costs by 2,000. Only material has 2,000 equivalent units.
- Valuing closing WIP at 500 × total rate 18 = 9,000. Wrong — you must value each element on **its own** equivalent units, giving 6,600.
- Skipping the statement of evaluation and jumping to the account. The three statements each carry marks of their own.

FINAL ANSWER

Cost per unit — Material / Labour / Overhead **₹ 9.00 / ₹ 5.00 / ₹ 4.00**

Total cost per unit **₹ 18.00**

Value of 1,500 units transferred **₹ 27,000**

Value of closing WIP (500 units) **₹ 6,600**

Process account total **₹ 33,600**

Proof: 18,000 + 9,000 + 6,600 = 33,600 = 27,000 + 6,600

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

During January 2,000 units were introduced into Process 1. The normal loss was estimated at 5% on input. At the end of the month 1,400 units had been produced and transferred to the next process. 460 units were incomplete and 140 units had been scrapped. It was estimated that incomplete units had reached a stage in production as follows: Material 75% complete, Labour 50% complete and overheads 50% complete.

The cost of 2,000 units introduced was ₹5,800. Direct material introduced during the process amounted to ₹1,440. Direct labour cost was ₹3,340. Production overheads incurred were ₹1,670. Units scrapped realised ₹1 each. The units scrapped have passed through the processes so were 100% complete as regards material, labour and overheads.

You are required to (a) prepare a statement of equivalent production (b) evaluate the cost of abnormal loss, finished goods and closing stock and (c) prepare the Process 1 a/c and abnormal loss a/c.

READ THE QUESTION FIRST — WHAT TO SPOT

- Units: 2,000 = 1,400 transferred + 460 WIP + 140 scrapped. Balanced
- **Normal loss = 5% of 2,000 = 100 units.** Actually scrapped = 140. So **abnormal loss = 40 units.**
- **Normal loss gets NIL equivalent units** in the statement. Abnormal loss gets 100% in every element, because the question tells you the scrapped units were fully processed.
- The scrap money of ₹1 per unit applies to the **normal loss units only** when you reduce the material cost. The abnormal loss units also fetch ₹1 each, but that goes in the Abnormal Loss account, not in the cost statement.

(a) Statement of Equivalent Production

Output / Particulars	Units	Material %	Material EU	Labour %	Labour EU	Overhead %	Overhead EU
Units completed & transferred	1,400	100	1,400	100	1,400	100	1,400
Normal loss (5% of 2,000)	100	—	Nil	—	Nil	—	Nil
Abnormal loss	40	100	40	100	40	100	40
Closing work-in-progress	460	75	345	50	230	50	230
Total equivalent units			1,785		1,670		1,670

(b1) Statement of Cost

Element	Cost ₹	Equivalent units	Cost per unit ₹
Material $5,800 + 1,440 - 100 \text{ scrap}$	7,140	1,785	4.00
Labour	3,340	1,670	2.00
Overhead	1,670	1,670	1.00
Total cost per unit			7.00

Material cost = cost of units introduced 5,800 + material added 1,440 = 7,240

Less: scrap value of **normal loss** 100 units × ₹1 = (100)

Net material cost = ₹7,140

(b2) Statement of Evaluation

Particulars	Working	Amount ₹
Units completed & transferred	1,400 × ₹7.00	9,800
Abnormal loss	40 × ₹7.00	280
Closing work-in-progress		
Material	345 EU × ₹4.00	1,380
Labour	230 EU × ₹2.00	460
Overhead	230 EU × ₹1.00	230
Value of closing WIP		2,070
Total		12,150

(c) Process 1 Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Units introduced	2,000	—	5,800	By Normal loss 100 units × ₹1	100	1	100
To Direct material	—	—	1,440	By Abnormal loss 40 × ₹7	40	7	280
To Direct labour	—	—	3,340	By Transfer to next process 1,400 × ₹7	1,400	7	9,800
To Production overheads	—	—	1,670	By Closing WIP statement (b2)	460	—	2,070
Total	2,000	—	12,250	Total	2,000	—	12,250

Abnormal Loss Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Process 1 A/c	40	7	280	By Cash / Cost Ledger (scrap) 40 × ₹1	40	1	40
Total	40	—	280	By Profit & Loss A/c	—	—	240
				Total	40	—	280

THE TRAP / WHERE MARKS ARE LOST

- **Giving the normal loss 100 equivalent units.** It gets nil. This is the single most common error in the topic.
- Deducting the scrap of all 140 scrapped units (140) from material cost instead of only the normal loss 100.
- Forgetting that the abnormal loss must be valued at the full rate of 7.00 even though it only sells for 1.00.

FINAL ANSWER

Equivalent units — Material / Labour / Overhead	1,785 / 1,670 / 1,670
Cost per unit — Material / Labour / Overhead	₹ 4.00 / ₹ 2.00 / ₹ 1.00
Total cost per unit	₹ 7.00
Finished goods (1,400 units)	₹ 9,800
Abnormal loss (40 units)	₹ 280 — net to P&L ₹ 240
Closing stock (460 units)	₹ 2,070

Proof: 12,250 debits = 100 + 280 + 9,800 + 2,070 = 12,250

Q19 Closing WIP has to be derived

Equivalent production · p.14

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

Prepare statement of equivalent production, statement of cost and process a/c from the following information.

Particulars	
Units introduced	7,600
Output (units)	6,000
Process cost — Materials	₹ 14,560
Process cost — Labour	₹ 21,360
Process cost — Overheads	₹ 14,240
Degree of completion for closing WIP — Materials	80%
Degree of completion for closing WIP — Labour	70%
Degree of completion for closing WIP — Overheads	70%

READ THE QUESTION FIRST — WHAT TO SPOT

- The closing WIP figure is **not given** — you must derive it: 7,600 introduced – 6,000 output = **1,600 units** of closing WIP.
- No losses are mentioned and the units balance exactly, so there is no normal or abnormal loss working at all.
- No opening WIP either, so the FIFO / average question does not arise.

Statement of Equivalent Production

Output / Particulars	Units	Material %	Material EU	Labour %	Labour EU	Overhead %	Overhead EU
Units completed & transferred	6,000	100	6,000	100	6,000	100	6,000
Closing WIP $7,600 - 6,000$	1,600	80	1,280	70	1,120	70	1,120
Total equivalent units			7,280		7,120		7,120

Statement of Cost

Element	Cost ₹	Equivalent units	Cost per unit ₹
Material	14,560	7,280	2.00
Labour	21,360	7,120	3.00
Overhead	14,240	7,120	2.00
Total cost per unit			7.00

Statement of Evaluation

Particulars	Working	Amount ₹
Units completed & transferred	$6,000 \times ₹7.00$	42,000
Closing work-in-progress		

Particulars	Working	Amount ₹
Material	1,280 EU × ₹2.00	2,560
Labour	1,120 EU × ₹3.00	3,360
Overhead	1,120 EU × ₹2.00	2,240
Value of closing WIP		8,160
Total		50,160

Process Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Materials	7,600	—	14,560	By Transfer to next process	6,000	7	42,000
To Labour	—	—	21,360	By Closing WIP	1,600	—	8,160
To Overheads	—	—	14,240				
Total	7,600	—	50,160	Total	7,600	—	50,160

FINAL ANSWER

Closing WIP (derived) **1,600 units**

Equivalent units — Material / Labour / Overhead **7,280 / 7,120 / 7,120**

Cost per unit **₹ 2.00 + ₹ 3.00 + ₹ 2.00 = ₹ 7.00**

Transferred out (6,000 units) **₹ 42,000**

Closing WIP **₹ 8,160**

Proof: 14,560 + 21,360 + 14,240 = 50,160 = 42,000 + 8,160

Q20 Normal loss, no abnormal item

Equivalent production · p.14

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

Prepare the process account from the following. Number of units introduced into the process — 4,000 units. Units completed and transferred — 3,000 units. Work-in-progress at the end of the period — 800 units (stage of completion material 80%, labour 70% and overheads 70%). Normal loss at the end of the period estimated 200 units (scrap value at ₹1 per unit). Value of materials ₹7,480, wages ₹10,680 and overheads ₹7,120.

READ THE QUESTION FIRST — WHAT TO SPOT

- Units: 4,000 = 3,000 transferred + 800 WIP + 200 normal loss. Balanced exactly, so **there is no abnormal loss or gain** — do not invent one.
- Normal loss again takes **nil equivalent units**, and its scrap value of 200 × ₹1 = ₹200 comes off the material cost.

Statement of Equivalent Production

Output / Particulars	Units	Material %	Material EU	Labour %	Labour EU	Overhead %	Overhead EU
Units completed & transferred	3,000	100	3,000	100	3,000	100	3,000
Normal loss	200	—	Nil	—	Nil	—	Nil

Output / Particulars	Units	Material %	Material EU	Labour %	Labour EU	Overhead %	Overhead EU
Closing WIP	800	80	640	70	560	70	560
Total equivalent units			3,640		3,560		3,560

Statement of Cost

Element	Cost ₹	Equivalent units	Cost per unit ₹
Material 7,480 – 200 scrap	7,280	3,640	2.00
Wages	10,680	3,560	3.00
Overhead	7,120	3,560	2.00
Total cost per unit			7.00

Statement of Evaluation

Particulars	Working	Amount ₹
Units completed & transferred	3,000 × ₹7.00	21,000
Closing work-in-progress		
Material	640 EU × ₹2.00	1,280
Wages	560 EU × ₹3.00	1,680
Overhead	560 EU × ₹2.00	1,120
Value of closing WIP		4,080
Total		25,080

Process Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Materials	4,000	—	7,480	By Normal loss 200 × ₹1	200	1	200
To Wages	—	—	10,680	By Transfer to next process	3,000	7	21,000
To Overheads	—	—	7,120	By Closing WIP	800	—	4,080
Total	4,000	—	25,280	Total	4,000	—	25,280

FINAL ANSWER

Equivalent units — Material / Wages / Overhead	3,640 / 3,560 / 3,560
Cost per unit	₹ 2.00 + ₹ 3.00 + ₹ 2.00 = ₹ 7.00
Transferred out (3,000 units)	₹ 21,000
Closing WIP (800 units)	₹ 4,080
Normal loss scrap	₹ 200
Proof: 25,280 = 200 + 21,000 + 4,080 = 25,280	

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

A company follows process costing and manufactures a product in one process. The work-in-process at the end of each month is valued according to **FIFO** method. At the beginning of the month of January, the inventory of WIP showed 400 units, 40% completed, valued as follows: Materials ₹3,600; Labour ₹3,400; Overheads ₹1,000 — total ₹8,000.

In the month of January, materials were purchased for ₹75,000. Wages and overheads in the month amounted to ₹79,800 and ₹21,280 respectively. **Actual issue of materials to production was ₹68,500.** Finished production taken into the stock in the month was 2,500 units. There was no loss in the process. At the end of the month, the WIP inventory was 500 units and 80% complete as regards materials and 60% complete as regards labour and overhead.

You are required to compute equivalent production and prepare process account.

READ THE QUESTION FIRST — WHAT TO SPOT

- **The ₹75,000 of purchases is a deliberate trap.** Purchases are not a process cost. Only what is **issued to production** — ₹68,500 — enters the process. Use 68,500 and ignore 75,000.
- **FIFO is stated explicitly.** So the opening WIP of 400 units, already 40% done, needs only the remaining **60%** of work this month → 240 equivalent units in each element.
- Units introduced must be derived: 2,500 completed + 500 closing WIP – 400 opening WIP = **2,600 units introduced.**
- Under FIFO the **opening WIP cost of ₹8,000 is kept out of the rate.** It is added back only when you value the finished goods.

WORKING NOTES

W1 Unit flow

Opening WIP = 400 units | Completed = 2,500 units | Closing WIP = 500 units

Units introduced = 2,500 + 500 – 400 = 2,600 units

Of the 2,500 completed, 400 came from opening WIP → **2,100 were introduced and completed this month**

Statement of Equivalent Production — FIFO

Output / Particulars	Units	Material %	Material EU	Labour %	Labour EU	Overhead %	Overhead EU
Opening WIP — work still to do 40% done, so 60% remains	400	60	240	60	240	60	240
Introduced and completed this month	2,100	100	2,100	100	2,100	100	2,100
Closing WIP	500	80	400	60	300	60	300
Total equivalent units			2,740		2,640		2,640

Statement of Cost — this month's costs only

Element	Cost ₹	Equivalent units	Cost per unit ₹
Material issued, not purchased	68,500	2,740	25.0000
Labour	79,800	2,640	30.2273

Element	Cost ₹	Equivalent units	Cost per unit ₹
Overhead	21,280	2,640	8.0606
Total cost per unit			63.2879

The labour and overhead rates do not come out round in this problem. Carry four decimal places through the evaluation and the process account will still balance to the rupee — which is shown below.

Statement of Evaluation

Particulars	Working	Amount ₹
1. Opening WIP completed (400 units)		
Cost brought forward	given	8,000.00
Material to finish	240 EU × ₹25.0000	6,000.00
Labour to finish	240 EU × ₹30.2273	7,254.55
Overhead to finish	240 EU × ₹8.0606	1,934.55
Total for the 400 units		23,189.10
2. Introduced and completed (2,100 units)		
At full cost	2,100 × ₹63.2879	1,32,904.55
3. Closing WIP (500 units)		
Material	400 EU × ₹25.0000	10,000.00
Labour	300 EU × ₹30.2273	9,068.18
Overhead	300 EU × ₹8.0606	2,418.18
Value of closing WIP		21,486.36
Total		1,77,580.01

Process Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Opening WIP	400	—	8,000	By Finished goods transferred 23,189.10 + 1,32,904.55	2,500	—	1,56,093.65
To Materials issued	2,600	—	68,500	By Closing WIP	500	—	21,486.36
To Wages	—	—	79,800				
To Overheads	—	—	21,280				
Total	3,000	—	1,77,580	Total	3,000	—	1,77,580.01

THE TRAP / WHERE MARKS ARE LOST

- Using the ₹75,000 **purchases** figure instead of the ₹68,500 **issued** figure. Purchases go to the stores account, not the process.
- Giving the opening WIP 400 equivalent units. Under FIFO it gets only the 240 that represent work done *this* month.
- Adding the opening WIP cost of ₹8,000 into the numerator before dividing. Under FIFO it stays out of the rate and is added back at evaluation.

FINAL ANSWER

Units introduced (derived)	2,600
Equivalent units — Material / Labour / Overhead	2,740 / 2,640 / 2,640
Cost per unit — Material	₹ 25.0000
Cost per unit — Labour	₹ 30.2273
Cost per unit — Overhead	₹ 8.0606
Finished goods (2,500 units)	₹ 1,56,093.65
Closing WIP (500 units)	₹ 21,486.36
Proof: $8,000 + 68,500 + 79,800 + 21,280 = 1,77,580$ and $1,56,093.65 + 21,486.36 = 1,77,580.01$ (1 paisa rounding).	

Q22 FIFO with scrap and two material inputs

Equivalent production · p.15

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

Prepare Process 2 account from the following. Opening stock 600 units at ₹1,050 (degree of completion material 80%, labour 60%, overheads 60%). Transfer from Process 1 — 11,000 units at ₹5,500. Transfer to Process 3 — 8,800 units. Direct material added in Process 2 — ₹2,410. Direct labour ₹7,155, Production overhead ₹9,540. Units scrapped — 1,200 (degree of completion material 100%, labour 70% and overheads 70%). Closing stock 1,600 units (degree of completion material 70%, labour 60%, overheads 60%). There was a normal loss in the process at 10% of production. Units scrapped would realise 50 paise per unit.

READ THE QUESTION FIRST — WHAT TO SPOT

- Units in: opening 600 + received 11,000 = 11,600. Units out: 8,800 + 1,200 scrapped + 1,600 closing = 11,600
- “Normal loss at 10% of production” is ambiguous wording. The standard reading is 10% of the units introduced into the process from Process 1 = 10% of 11,000 = **1,100 units**. State your assumption in one line. Abnormal loss then = $1,200 - 1,100 = 100$ units.
- The opening stock’s degrees of completion are given element by element, so **use FIFO**.
- Two material figures must be added: the ₹5,500 transferred in and the ₹2,410 added here. Then take off the normal loss scrap of $1,100 \times ₹0.50 = ₹550$.

WORKING NOTES

W1 Loss analysis (assumption stated)

Normal loss = 10% of 11,000 units introduced = **1,100 units**

Units actually scrapped = 1,200 units

Abnormal loss = $1,200 - 1,100 = 100$ units

Scrap credit taken to cost statement = $1,100 \times ₹0.50 = ₹550$

W2 FIFO unit flow

Opening stock 600 units, so of the 8,800 transferred out, 600 came from opening stock

Introduced and completed = $8,800 - 600 = 8,200$ units

Statement of Equivalent Production — FIFO

Output / Particulars	Units	Material %	Material EU	Labour %	Labour EU	Overhead %	Overhead EU
	600	20	120	40	240	40	240

Output / Particulars	Units	Material %	Material EU	Labour %	Labour EU	Overhead %	Overhead EU
Opening stock — work still to do Mat 20%, Lab 40%, OH 40%							
Introduced and completed	8,200	100	8,200	100	8,200	100	8,200
Normal loss (W1)	1,100	—	Nil	—	Nil	—	Nil
Abnormal loss (W1)	100	100	100	70	70	70	70
Closing stock	1,600	70	1,120	60	960	60	960
Total equivalent units			9,540		9,470		9,470

Statement of Cost

Element	Cost ₹	Equivalent units	Cost per unit ₹
Material $5,500 + 2,410 - 550$	7,360	9,540	0.7715
Labour	7,155	9,470	0.7555
Overhead	9,540	9,470	1.0074
Total cost per unit			2.5344

Statement of Evaluation

Particulars	Working	Amount ₹
1. Opening stock completed (600 units)		
Cost brought forward	given	1,050.00
Material to finish	$120 \text{ EU} \times ₹0.7715$	92.58
Labour to finish	$240 \text{ EU} \times ₹0.7555$	181.33
Overhead to finish	$240 \text{ EU} \times ₹1.0074$	241.77
Sub-total		1,565.68
2. Introduced and completed (8,200 units)		
At full cost	$8,200 \times ₹2.5344$	20,782.28
Transferred to Process 3 (8,800 units)		22,347.96
3. Abnormal loss (100 units)		
Material / Labour / Overhead	$100 \times 0.7715 + 70 \times 0.7555 + 70 \times 1.0074$	200.56
4. Closing stock (1,600 units)		
Material	$1,120 \text{ EU} \times ₹0.7715$	864.07
Labour	$960 \text{ EU} \times ₹0.7555$	725.32
Overhead	$960 \text{ EU} \times ₹1.0074$	967.10
Value of closing stock		2,556.49
Total		25,105.01

Process 2 Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Opening stock	600	—	1,050	By Normal loss $1,100 \times ₹0.50$	1,100	0.50	550
To Transfer from Process 1	11,000	0.50	5,500	By Abnormal loss	100	—	200.56
To Direct material added	—	—	2,410	By Transfer to Process 3	8,800	—	22,347.96
To Direct labour	—	—	7,155	By Closing stock	1,600	—	2,556.49
To Production overhead	—	—	9,540				
Total	11,600	—	25,655	Total	11,600	—	25,655.01

THE TRAP / WHERE MARKS ARE LOST

About the ambiguous wording. “Normal loss at 10% of production” could be read as 10% of units introduced (1,100), 10% of total input including opening stock (1,160), or 10% of output transferred (880). I have used **10% of units introduced = 1,100**, which is the most common convention. In the exam, **write down which basis you have used** in one line. Method marks are given for a stated, consistent assumption; they are lost only for silence.

FINAL ANSWER

Normal loss / Abnormal loss	1,100 units / 100 units
Equivalent units — Material / Labour / Overhead	9,540 / 9,470 / 9,470
Cost per unit	₹ 0.7715 + ₹ 0.7555 + ₹ 1.0074 = ₹ 2.5344
Transferred to Process 3 (8,800 units)	₹ 22,347.96
Abnormal loss (100 units)	₹ 200.56
Closing stock (1,600 units)	₹ 2,556.49

Proof: debits 1,050 + 5,500 + 2,410 + 7,155 + 9,540 = 25,655 and credits 550 + 200.56 + 22,347.96 + 2,556.49 = 25,655.01

Q23 The average method

Equivalent production · p.15-16

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

The following figures relate to a single industrial process. Opening stock 10,000 units — Material ₹2,250, Wages ₹650, Overheads ₹400, total ₹3,300. Units introduced 40,000 units — Material ₹9,250, Labour ₹4,600, Overheads ₹3,100.

During the period 30,000 units were completed and 20,000 units remained in process. The degree of completion of closing stock of WIP was: Materials 100%, Labour 25%, Overheads 25%. Make necessary computations and prepare process account by using the **average method**.

READ THE QUESTION FIRST — WHAT TO SPOT

- **The question says average method**, so everything changes from Q21 and Q22:
- The opening stock does **not** appear as a separate line in the equivalent production statement. Its 10,000 units are simply inside the 30,000 completed, at 100%.
- The opening stock **cost is merged** with this month's cost before you divide. Material becomes 2,250 + 9,250 = 11,500, and so on.
- Units: 10,000 + 40,000 = 50,000 = 30,000 completed + 20,000 WIP no losses.

Statement of Equivalent Production — Average method

Output / Particulars	Units	Material %	Material EU	Labour %	Labour EU	Overhead %	Overhead EU
Units completed & transferred <i>includes the opening 10,000 at 100%</i>	30,000	100	30,000	100	30,000	100	30,000
Closing WIP	20,000	100	20,000	25	5,000	25	5,000
Total equivalent units			50,000		35,000		35,000

Statement of Cost — opening cost merged with current cost

Element	Cost ₹	Equivalent units	Cost per unit ₹
Material <i>2,250 + 9,250</i>	11,500	50,000	0.23
Wages <i>650 + 4,600</i>	5,250	35,000	0.15
Overheads <i>400 + 3,100</i>	3,500	35,000	0.10
Total cost per unit			0.48

Statement of Evaluation

Particulars	Working	Amount ₹
Units completed & transferred	30,000 × ₹0.48	14,400
Closing work-in-progress		
Material	20,000 EU × ₹0.23	4,600
Wages	5,000 EU × ₹0.15	750
Overheads	5,000 EU × ₹0.10	500
Value of closing WIP		5,850
Total		20,250

Process Account (average method)

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Opening stock	10,000	—	3,300	By Transfer to finished stock	30,000	0.48	14,400
To Material	40,000	—	9,250	By Closing WIP	20,000	—	5,850
To Labour	—	—	4,600				
To Overheads	—	—	3,100				
Total	50,000	—	20,250	Total	50,000	—	20,250

WHY THIS WORKS

Notice how much shorter this is than Q21. Under the average method you never separate “work done last month” from “work done this month” — you pool everything and take one average rate. That is why it is the easier method, and why examiners often specify FIFO instead.

FINAL ANSWER

Equivalent units — Material / Wages / Overheads **50,000 / 35,000 / 35,000**

Cost per unit **₹ 0.23 + ₹ 0.15 + ₹ 0.10 = ₹ 0.48**

Transferred out (30,000 units) **₹ 14,400**

Closing WIP (20,000 units)

₹ 5,850

Proof: $3,300 + 9,250 + 4,600 + 3,100 = 20,250 = 14,400 + 5,850$

Q24 FIFO and average on the same data

Equivalent production · p.16

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

Prepare the statement of equivalent production from the following under **FIFO and Average Method**.

Opening work-in-progress — 2,000 units (completed as material 80%, labour 60% and overheads 60%). Units introduced and completed 5,000 units. Units remaining as closing work-in-progress — 3,000 units (completed as material 80%, labour and overheads 60% each). There are no process losses.

READ THE QUESTION FIRST — WHAT TO SPOT

- No costs are given, so **only the two statements are wanted** — no cost statement, no evaluation, no process account. Answer exactly what is asked.
- Total units completed = opening WIP 2,000 (finished first) + 5,000 introduced and completed = **7,000 units**. Units introduced = 5,000 + 3,000 closing = **8,000 units**.
- This is the best problem in the module for **seeing** the FIFO/average difference, because the same data gives two different answers. Work both and compare the totals.

(a) Statement of Equivalent Production — FIFO method

Output / Particulars	Units	Material %	Material EU	Labour %	Labour EU	Overhead %	Overhead EU
Opening WIP — work still to do Mat 20%, Lab 40%, OH 40%	2,000	20	400	40	800	40	800
Introduced and completed	5,000	100	5,000	100	5,000	100	5,000
Closing WIP	3,000	80	2,400	60	1,800	60	1,800
Total equivalent units			7,800		7,600		7,600

(b) Statement of Equivalent Production — Average method

Output / Particulars	Units	Material %	Material EU	Labour %	Labour EU	Overhead %	Overhead EU
Units completed & transferred 2,000 opening + 5,000 = 7,000, all at 100%	7,000	100	7,000	100	7,000	100	7,000
Closing WIP	3,000	80	2,400	60	1,800	60	1,800
Total equivalent units			9,400		8,800		8,800

The difference, side by side

Element	FIFO equivalent units	Average equivalent units	Difference	Why
Material	7,800	9,400	1,600	Average counts the opening WIP's material at 100% (2,000) instead of only the 20% still to do (400). Difference = $2,000 - 400 = 1,600$.
Labour	7,600	8,800	1,200	2,000 at 100% instead of 800 → difference 1,200.

Element	FIFO equivalent units	Average equivalent units	Difference	Why
Overhead	7,600	8,800	1,200	Same as labour.

WHY THIS WORKS

Average always gives the **larger** number of equivalent units, because it counts the whole of the opening WIP instead of just the unfinished part. Larger equivalent units with a larger cost pool (opening cost is merged in too) means the two methods usually give similar rates — but never identical ones. Whenever a question mentions opening WIP, your first job is to find out which method it wants.

FINAL ANSWER

FIFO — Material / Labour / Overhead **7,800 / 7,600 / 7,600**

Average — Material / Labour / Overhead **9,400 / 8,800 / 8,800**

Units completed **7,000**

Units introduced **8,000**

Under FIFO only the 400 / 800 / 800 of opening-WIP work belongs to this period; under average the full 2,000 units are counted at 100%.